Chair of Precision Measurements at Extreme Conditions
Department of Physics
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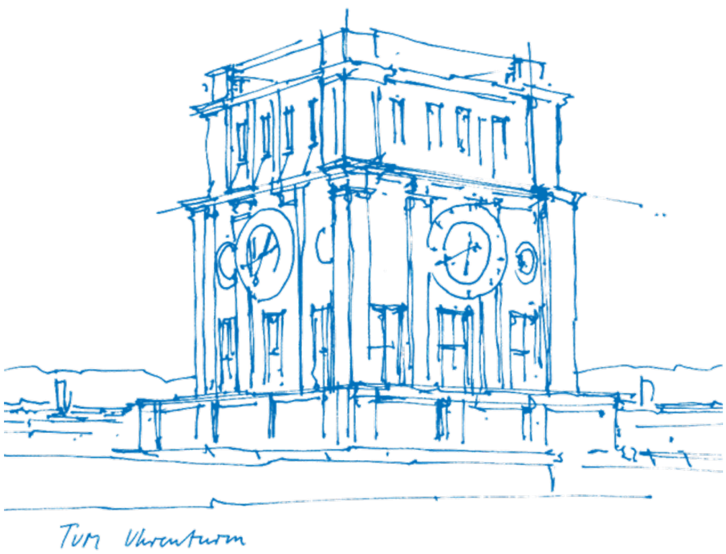
Magnetocardiographic Screening of Adults with Arrhythmogenic Cardiomyopathy

A Machine Learning-Enhanced Approach

Samuel Clemens Friese



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Samuel Clemens Friese

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I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

Munich, 11.08.2025 Samuel Clemens Friese

Abstract

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Arrhythmogenic Cardiomyopathy (ACM) is an inherited heart muscle disease characterized by a high risk of sudden cardiac death, yet its diagnosis remains challenging. Magnetocardiography (MCG), which measures the magnetic fields generated by the heart's electrical activity, offers a non-invasive and highly sensitive alternative to conventional electrocardiography. However, the clinical application of MCG relies on robust and precise signal processing methods. This thesis presents a machine learning-enhanced approach to investigate the potential of MCG for screening ACM.

A deep neural network based on a U-Net architecture was developed and trained for the automated segmentation of cardiac waveforms into their constituent P-wave, QRS complex, and T-wave components. This model was specifically designed to operate on noisy, short-interval signals characteristic of multi-channel MCG recordings. Subsequently, a pilot study was conducted using an innovative Optically Pumped Magnetometer (OPM) system to record MCG data from a cohort of 21 participants (9 with a clinical diagnosis of ACM and 12 healthy controls). The trained segmentation model was deployed to delineate cardiac intervals, enabling a detailed analysis of the *Vector Magnetocardiogram* (VMCG) loop dynamics.

The machine learning model achieved high-performance waveform segmentation, providing a robust foundation for VMCG analysis. The clinical study revealed statistically significant differences in the VMCG of ACM patients compared to healthy controls. These differences were most prominent and widespread in the T-wave segment, where both the VMCG loop area and maximum amplitude were significantly reduced in the ACM cohort. These effects were spatially concentrated in the lower and right-sided sensor regions, corresponding anatomically to the ventricles. Significant signal attenuation was also observed in the QRS complex, particularly in the aggregated sub-grid analysis. In contrast, no significant differences were observed in the ST segment, and shape descriptors (Solidity and Compactness) remained stable between groups. These findings suggest that the primary electrophysiological abnormalities manifest as a coherent loss of signal strength, particularly during ventricular repolarization, establishing VMCG parameters as promising biomarkers for the non-invasive screening of ACM.

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1 Introduction

The electrophysiological activity of the human organs is of great scientific and clinical interest. While several techniques exist, this work focuses on biomagnetism: the measurement of weak but information-rich magnetic fields generated by the body. Significant areas of application include magnetocardiography (MCG) for the heart, magnetoencephalography (MEG) for the brain, and magnetopneumography (MPG) for the lungs [44].

MCG in particular offers a non-invasive and risk-free technique for recording the magnetic fields induced by the heart's electrical activity. It possesses significant diagnostic potential, particularly for localizing electrophysiologic phenomena within the heart [6], providing spatially and temporally accurate measurements of the extremely weak magnetic fields (on the order of 10^{-13} to 10^{-12} T), which are almost a million times weaker than the Earth's magnetic field. Both MCG and the more common electrocardiography (ECG) originate from the same electrophysiological sources. Cardiac contraction, triggered by the depolarization of myocardial cells, is associated with electrical phenomena that create a faint magnetic field detectable on the chest surface. This results in both signals showing similar features, such as the P-wave, the QRS complex, and the T-wave (see Figure 1.1) [36].

A Detailed Comparison: MCG vs. ECG

MCG presents several significant advantages over ECG. The core difference lies in the sources they measure. MCG directly measures the magnetic field, a three-dimensional vector, generated by the *impressed or primary currents* ($\boldsymbol{J_i}$). These currents consist of active ion flows across the cell membranes of myocardial fibers and represent the fundamental source of cardiac electrophysiology. In contrast, the ECG measures scalar potential differences between electrodes on the skin, which are generated by passive, *extracellular secondary or volume currents* ($\boldsymbol{J_v}$). These volume currents are driven through the conductive torso by the electric fields created by the primary currents [36].

Consequently, the electric signal is significantly attenuated and smeared by the highly variable and often poorly characterized electrical conductivity (σ) of different tissue layers like skin, fat, and bone [36]. The magnetic signal, however, passes through these tissues virtually undistorted because their magnetic permeability (μ) is almost identical to that of free space. Furthermore, modern multi-channel MCG systems enable the simultaneous acquisition of a complete magnetic field map over the chest in a single recording, eliminating the need for sequential repositioning of sensors and providing rich spatial information [44].

The Technological Evolution of MCG Sensors

The first single-channel MCG was recorded in 1963 by Baule and McFee using large copper induction coils configured as a gradiometer to cancel overwhelming ambient magnetic noise. However, these early systems were constrained by a poor signal-to-noise ratio, which limited significant progress [36]. The advent of superconductivity, marked by the publication of BCS theory and the theoretical description of the Josephson junction, led to the development of Superconducting Quantum Interference Devices (SQUIDs) [9]. With their exceptionally high sensitivity, SQUIDs became the gold standard for biomagnetic measurements. Their primary drawback, however, is the requirement for cryogenic cooling with liquid helium, which makes SQUID-based systems expensive, bulky, and inflexible. This has hindered their widespread clinical adoption, as in MCG, the sensors are required to be placed in close proximity to the human torso [36].

More recently, a new generation of sensors has emerged as a compelling alternative: Optically Pumped Magnetometers (OPMs). OPMs offer a sensitivity comparable to SQUIDs but operate at room temperature. As noted in recent studies, this technology has led to compact, less expensive sensors that can be flexibly arranged to conform to a patient's anatomy and operate within smaller, person-sized magnetic shields rather than entire magnetically shielded rooms. This thesis utilizes an advanced OPM-based system similar to the one proposed in Wurm et al. [43], which is discussed further in Chapters 2.2 and 4.1.

The Challenge of the Inverse Problem

A common goal in analyzing MCG data is to reconstruct the underlying cardiac current distribution ($\mathbf{J}$), a task known as the *inverse problem*. A successful reconstruction would offer a deep understanding of abnormalities and their precise locations. However, this problem is notoriously ill-posed due to the fundamental nature of magnetostatics. The magnetic field ($\mathbf{B}$) is generated by the source current ($\mathbf{J}$) according to the Biot-Savart law [14]:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{r}') \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} \, d^3r' \tag{1.1}$$

The physical origin of the inverse problem's ambiguity becomes clear by reformulating this law using a vector identity, which separates the field's sources into two distinct types [21]:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_S \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{n}}}{|\mathbf{r} - \mathbf{r}'|} \, d^2r' + \frac{\mu_0}{4\pi} \int_V \frac{\nabla' \times \mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, d^3r' \tag{1.2}$$

This expression reveals that the magnetic field has two sources. The first integral represents the field generated by currents flowing along surfaces (S) where conductivity changes (e.g., the heart-torso boundary). The second integral represents the field generated by "vortex" or rotational currents within the volume (V), characterized by a non-zero curl ($\nabla \times \mathbf{J} \neq 0$) [21].

A current distribution is *"magnetically silent"* if it produces no external magnetic field, meaning $\mathbf{B}(\mathbf{r}) = 0$. For this to occur, both integrals in Equation 1.2 must vanish. The volume integral is zero if the current is curl-free ($\nabla \times \mathbf{J} = 0$) everywhere. The surface integral is zero if the current is fully contained within a region of uniform conductivity and is zero at its boundaries. This has a critical physical implication: any curl-free current distribution that is confined to a single tissue type (like the myocardium) and does not reach a boundary is completely invisible to external magnetic field sensors. Since an infinite number of such *"silent sources"* can be added to a true current distribution without changing the measured MCG, the inverse problem is fundamentally non-unique. Consequently, a solution can only be found by applying significant physiological and anatomical constraints, often by iteratively solving the *forward problem* [21].

An Alternative Approach: Vector Magnetocardiography (VMCG)

This thesis explores an alternative, data-driven approach that bypasses the complexities of the inverse problem. The method is centered on *Vector Magnetocardiography (VMCG)*, defined as the trajectory traced by the magnetic field vector's tip over a specific cardiac cycle interval, such as the QRS complex or T-wave (see Figure 1.1). This trajectory can be projected onto different planes (e.g., the yz- and xy-planes) to create two-dimensional vector loops for analysis. The VMCG thus captures the time evolution of the magnetic field vector and is simultaneously highly dependent on sensor placement with respect to the patient's heart [44].

The diagnostic power of this concept has been established. For instance, a foundational study by Brala et al. (2023) [4] demonstrated that analyzing a magnetocardiography vector could effectively screen for inflammatory cardiomyopathy. The authors identified a vector-derived threshold that differentiated patients from healthy controls with high specificity (95%) and positive predictive value (93%). Crucially, they also

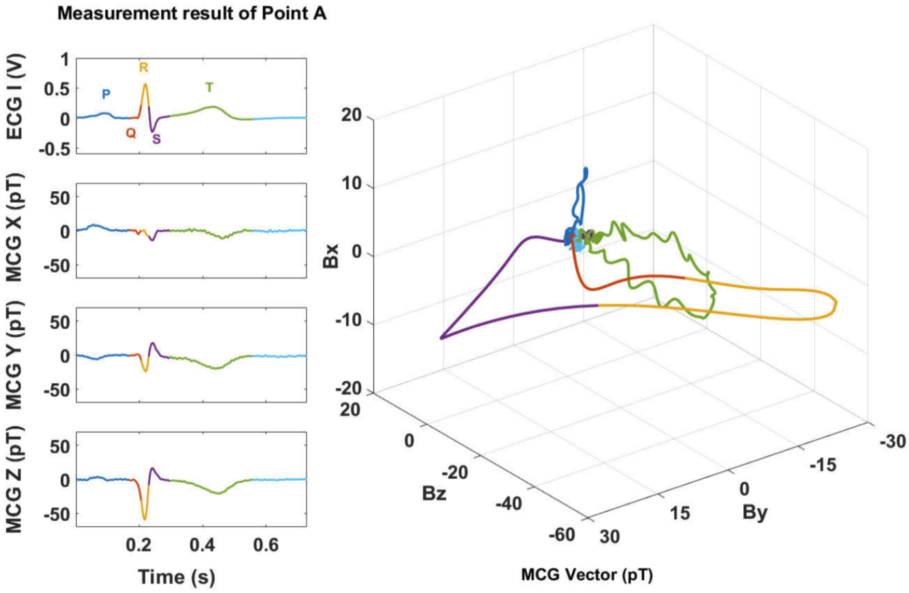


Figure 1.1 The left side shows the ECG signal and the MCG signals in different directions. For the sake of distinction, P-, QRS-, and T-waves are marked with different colors. The right side is the corresponding VMCG map [44, Fig. 7].

showed that the vector's normed area decreased and its orientation changed in response to successful immunosuppressive therapy, highlighting its value for monitoring treatment [4].

This thesis aims to apply and extend this VMCG concept to a different pathology: Arrhythmogenic Cardiomyopathy (ACM, formerly Arrhythmogenic Right Ventricular Cardiomyopathy, ARVC).

A pilot study was conducted on 22 participants (10 ACM-positive, 12 healthy controls) using an innovative OPM-based system at the Deutsches Herzzentrum München [43]. To facilitate the analysis, a neural network was developed and trained to automatically segment the MCG signal into its P-wave, QRS complex, and T-wave components, enabling precise analysis of the vector loops. The training process will be highlighted in Chapter 3. Chapter 2 will provide a more detailed background on the underlying concepts, including OPM physics and the pathophysiology of ACM, while Chapter 4 will discuss the experimental setup and results in detail.

2 Theoretical Background

2.1 Arrhythmogenic Cardiomyopathy

The term *Cardiomyopathy* was formally defined in 1995 by the World Health Organisation (WHO) as: *“diseases of the myocardium associated with cardiac dysfunction”* [27], with *myocardium* referring to the thick muscular middle layer of the heart wall, as illustrated in Figure 2.1.

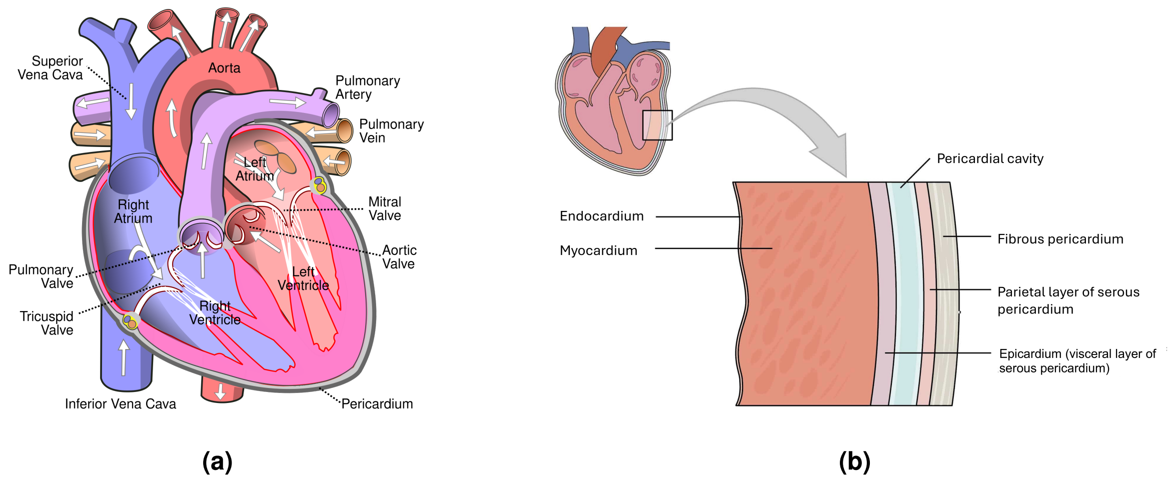


Figure 2.1 (a) Diagram of the human heart showing anatomical features such as the ventricles, atria, valves, and great vessels [42]. **(b)** Cross-sectional view of the heart wall illustrating its three main layers: endocardium (inner), myocardium (middle), and epicardium (outer) [37].

Before this definition was established, the term was often used to describe *“heart muscle diseases of unknown cause”* and was differentiated from *specific heart muscle disease* (of known cause) [27].

The WHO further differentiates between five major subclasses [27]:

Dilated Cardiomyopathy (DCM): Characterized by the enlargement (dilation) of one or both ventricles and impaired systolic function, which reduces the heart’s ability to pump blood effectively. It is the most common form of cardiomyopathy.

Hypertrophic Cardiomyopathy (HCM): Defined by an abnormal thickening (hypertrophy) of the heart muscle, particularly the interventricular septum, which is not explained by other causes like hypertension. This thickening can obstruct blood flow and impair ventricular relaxation.

Restrictive Cardiomyopathy (RCM): Marked by stiff and noncompliant ventricular walls that resist normal filling with blood between heartbeats (diastolic dysfunction). While the heart’s pumping (systolic) function may remain relatively normal, the impaired filling can lead to heart failure.

Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC): A hereditary condition where the muscle tissue of the right ventricle is progressively replaced by fatty and fibrous tissue. This disruption leads to life-threatening ventricular arrhythmias and structural abnormalities.

Unclassified Cardiomyopathies: A category for rare or mixed forms of cardiomyopathy that do not fit neatly into the other classifications, such as fibroelastosis or non-compacted myocardium.

The term *Right Ventricular Cardiomyopathy* was first established by Thiene et al. [38], who conducted postmortem studies on young individuals who had died suddenly in the Veneto region of Italy. Their finding that ARVC was the cause in 20% of these cases highlighted that the disease was more prevalent than previously thought, particularly in that region.

While historically defined by its impact on the right ventricle, it is now clear that the disease spectrum is much broader. Thanks to advances in genetics and imaging, variants with significant or even dominant involvement of the left ventricle (LV) are now well-recognized [30]. This has led to a proposed renaming from ARVC to the more inclusive term *Arrhythmogenic Cardiomyopathy (ACM)*, which better reflects the biventricular nature of the disease. A modern definition describes ACM as:

> *"AC is characterized by progressive fibro-fatty replacement of ventricular myocardium, including RV and LV, with relative sparing of the septum"* [30, p. 2].

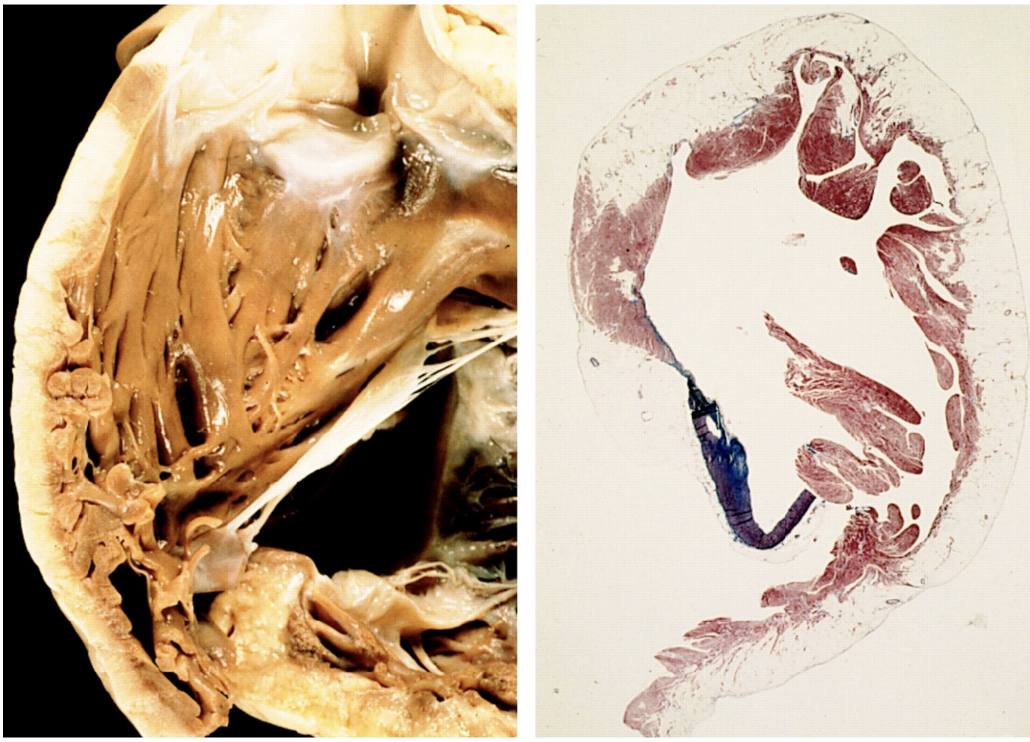
The histological features of this fibro-fatty replacement are shown in Figure 2.2.

ACM is typically diagnosed between the 20th and 40th year of life, although cases in older individuals are occasionally reported. Sudden cardiac arrest is oftentimes the first presentation of the disease. If prior symptoms do occur, they are usually in the form of arrhythmias (e.g., abnormal heart palpitations) and effort-induced syncope [5, 30].

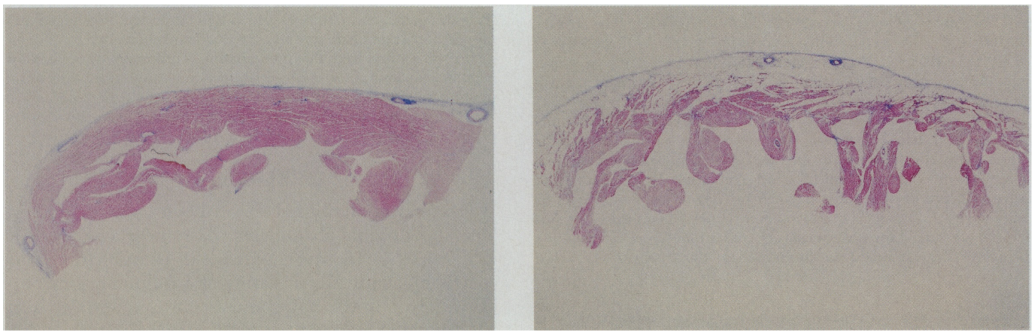
The genetic basis of ACM is strongly linked to mutations in genes encoding *desmosomal proteins*. Desmosomes are intercellular junctions that provide strong adhesion between cells, especially in tissues under significant mechanical stress like the heart and skin. In fact, this genetic link was first drawn from studies of Naxos disease, a condition characterized by a combination of ACM, skin abnormalities (palmoplantar keratosis), and hair abnormalities (woolly hair) [5]. While mutations observed in Naxos disease are one cause, mutations in other desmosomal genes (11 have been identified in total) are more common in the general ACM population [30]. The leading hypothesis, supported by ultrastructural studies of myocardial biopsies, is that these genetic defects create weak intercellular junctions. This weakness can lead to myocyte (muscle cell) detachment, cell death, and subsequent replacement with fibrofatty tissue. It is believed that physical exercise, which increases mechanical stress on the heart's wall, may promote this process [5]. Furthermore, some studies suggest that high-level endurance exercise can lead to high strain on the right ventricle, causing repetitive injury that results in pathological changes resembling those of ACM. This phenomenon is often called 'exercise-induced arrhythmogenic RV cardiomyopathy' [10], although potential overlaps with the well-documented "athlete's heart" exist [30].

The prevalence of ACM is estimated to range from 1 in 250 to 1 in 5,000 people globally [29, 5, 30], with higher rates reported in certain regions such as Northern Italy and Germany, where it may reach up to 1 in 2,000 [5, 30]. This makes ACM one of the more common among rare diseases [29]. In fact, ACM has been reported as the second leading cause of sudden death (SD) in the young and the primary cause in competitive athletes in the Veneto Region of Italy [30, 38]. About half of individuals with this condition have affected family members, as shown in the pedigree analysis in Figure 2.3. The disease is usually inherited in an autosomal dominant pattern, meaning only one copy of the faulty gene is needed. However, not everyone who inherits the gene shows symptoms (a concept known as incomplete penetrance). In rare cases, an autosomal recessive form exists. The disease tends to be more severe and occur more often in men than in women, with a ratio of about 3-to-1. This disparity may be due to hormonal effects or differences in exercise intensity between sexes [5].

There is no single definitive test for Arrhythmogenic Cardiomyopathy. Instead, diagnosis relies on a combination of findings, including heart structure and function (from echocardiography or MRI), tissue analysis (biopsy), ECG results, arrhythmias, and family history [30]. Specific ECG patterns, a confirmed ACM diagnosis in a close relative, or the identification of a known disease-causing mutation are considered major diagnostic criteria. However, caution is advised when interpreting genetic results, as the role of some mutations is still debated. Diagnosis remains challenging because ECG changes can be non-specific, various conditions can cause right ventricular arrhythmias, and imaging the right ventricle is difficult.



(a) Left: A 39-year-old man with a family history of sudden death and irregular heart rhythms died unexpectedly while resting. The heart showed significant fatty tissue buildup in the outer wall of the right ventricle. Right: Microscopic examination of the right ventricle revealed extensive fatty tissue replacing most of the myocardium, reaching the endocardium, with no scar tissue present [1, Fig. 7].



(b) Left: Microscopic view of heart tissue from a 14-year-old girl who died of leukemia. Right: Corresponding section from a heart exhibiting morphological features of arrhythmogenic right ventricular cardiomyopathy (ARVC/ACM) [38, Fig. 1].

Figure 2.2 Clinical and histological features associated with arrhythmogenic cardiomyopathy.

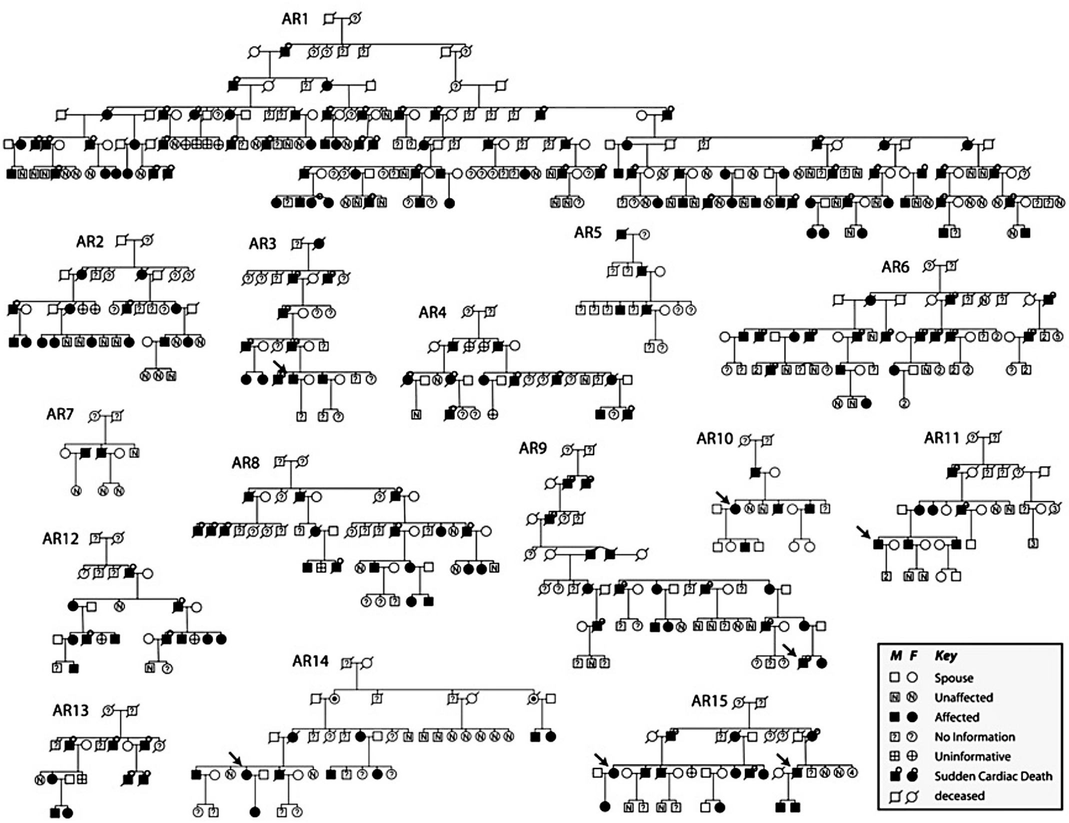


Figure 2.3 Family trees of 15 Newfoundland-ancestral families linked to Arrhythmogenic Right Ventricular Cardiomyopathy/Dysplasia Type 5 (ARVC Type 5 / ARVD5). Affected individuals are shown as filled-in squares (men) or circles (women). Those who died suddenly due to sudden cardiac arrest are represented by a small circle above their symbol [25, Fig. 1a].

Clinicians often face challenges in distinguishing ACM from other conditions with similar signs, such as myocarditis, sarcoidosis, and right ventricular heart attack [5, 30].

The main goals of clinical therapy for ACM are to reduce the risk of sudden death and to improve quality of life by managing arrhythmias and symptoms of heart failure, as outlined in Figure 2.4. A lifelong restriction from intense physical activity is advised for symptomatic individuals and even for healthy gene carriers, for whom this is often the only intervention needed alongside regular monitoring. While strong evidence is limited, beta-blockers are currently recommended for all symptomatic patients to slow the heart rate and reduce stress on the heart. For patients with severe, treatment-resistant heart failure or uncontrollable ventricular arrhythmias, heart transplantation may be considered [5, 30].

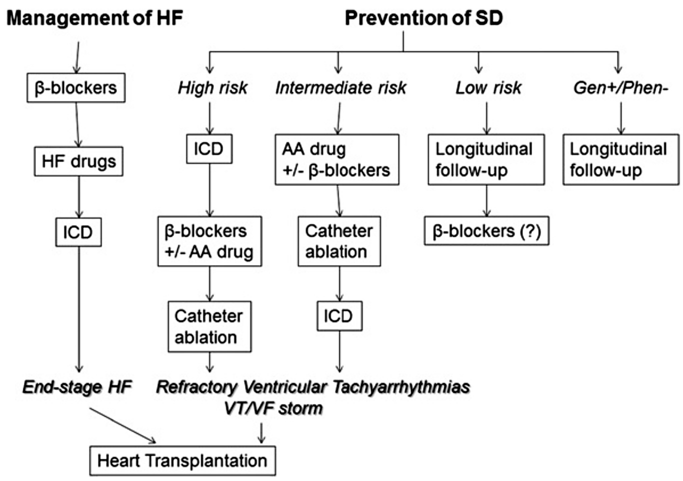


Figure 2.4 Flowchart outlining the clinical treatment of patients with arrhythmogenic cardiomyopathy (ACM). The left side shows the management of heart failure (HF), while the right side addresses the management of ventricular arrhythmias and sudden death (SD) prevention based on SD risk stratification [30, Fig. 8].

The risk of sudden death in ACM varies but is generally low with treatment, less than 1% per year in recent studies. Prognosis depends mainly on the severity of arrhythmias and heart muscle weakness, with a history of serious arrhythmic events increasing the risk of life-threatening complications [5].

2.2 Optically Pumped Magnetometers and the Measurement Setup

An Optically Pumped Magnetometer (OPM) operates by using light to prepare and probe a magnetically sensitive atomic vapor. In a typical configuration, a laser emits circularly polarized light that passes through a sealed vapor cell containing an alkali element, such as Rubidium (^{87}Rb). The light drives a process known as *optical pumping*, which prepares the atomic ensemble in a highly polarized state. The subsequent interaction of this polarized vapor with an external magnetic field alters the light's transmission, which is then measured by a photodetector, such as a photodiode. While some setups use a single laser for both pumping and probing, more complex configurations may employ a dedicated pump beam and a separate probe beam. In either case, the resulting intensity signal at the photodiode is proportional to the magnetic-field-dependent state of the vapor [39].

To understand how this magnetically sensitive state is created, one must first consider the atomic energy level structure. The primary energy levels of an atom (E_n) are split into a "fine structure" due to the coupling between the electron's orbital angular momentum ($\vec{L}$) and its spin angular momentum ($\vec{S}$), along with relativistic corrections. These fine structure levels are further split into "hyperfine" levels by the interaction of the electron's total angular momentum ($\vec{J}$) with the nuclear spin ($\vec{I}$). The possible hyperfine states are characterized by the total atomic angular momentum quantum number, F, which can take values in the range:

$$|I - J| \leq F \leq I + J \tag{2.1}$$

A specific hyperfine state is thus fully described by the ket $|n, F, J, I, m_F\rangle$. Here, m_F is the magnetic quantum number corresponding to F, representing the eigenvalue of the projection of the total angular momentum operator onto the quantization axis (F_z). The quantum number m_F can take any of the $2F + 1$ integer values from $-F$ to $+F$. Without loss of generality, this z-axis is defined to be parallel to the laser's propagation direction [39, 32].

In a weak external magnetic field, the initially degenerate hyperfine levels split into distinct Zeeman sublevels based on their m_F value. Transitions between these energy levels are not arbitrary but are governed by selection rules. The probability of a transition between an initial state $|i\rangle$ and a final state $|f\rangle$ is given by Fermi's Golden Rule, which expresses the transition rate $W_{i\to f}$ as:

$$W_{i\to f} = \frac{2\pi}{\hbar} \left|\langle f \,|\, H' \,|\, i\rangle\right|^2 \rho(E_f) \tag{2.2}$$

Here, $\rho(E_f)$ is the density of final states, and $\langle f \,|\, H' \,|\, i\rangle$ is the transition matrix element, which quantifies how strongly the two states are coupled by the perturbation H'. In this case, the perturbation is the oscillating electromagnetic field of the laser light [32].

For transitions induced by light, we can apply the *electric dipole approximation*, valid when the light's wavelength is much larger than the atom's size. The transition matrix element then becomes:

$$M_{if} = \langle f \,|\, H' \,|\, i\rangle \approx -\vec{e} \cdot \left\langle f \,\middle|\, \vec{d} \,\middle|\, i \right\rangle \tag{2.3}$$

where $\vec{d}$ is the electric dipole operator and $\vec{e}$ is the polarization vector of the light. If this matrix element is zero, the transition is *electric-dipole forbidden*. For circularly polarized light, conservation of angular momentum requires the photon's spin angular momentum to transfer to the atom. This leads to a strict selection rule for the magnetic quantum number: $\Delta m_F = \pm 1$ for right- or left-hand circularly polarized light, respectively [39, 32].

This principle is exploited to accumulate atoms in a non-absorbing quantum state, known as a "dark state." While the specific dark state depends on the laser polarization and atomic species, a common implementation (e.g., in the QuSpin sensors [28]), illustrative of the general mechanism, involves an ensemble of ^{87}Rb atoms. When illuminated by left-circularly polarized light that drives transitions with $\Delta m_F = +1$ (see Figure 2.5), the process unfolds as follows

1. **Absorption:** An atom absorbs a photon, causing its m_F value to increase by one.
2. **Spontaneous Emission:** The atom then relaxes back to a ground state by spontaneously emitting a photon. This emission is isotropic, meaning the change in m_F is probabilistic: Δm_F can be $0, +1$, or -1 with roughly equal likelihood.

Over many absorption-emission cycles, the deterministic absorption ($\Delta m_F = +1$) and randomizing emission result in a net drift of the atomic population towards states with higher m_F values. Atoms accumulate in the highest possible Zeeman sublevel (e.g., the $m_F = 2$ state for the $F = 2$ level), where they become "trapped" in a *dark state*. An atom in this state cannot absorb any more light because no available higher energy level satisfies the $\Delta m_F = +1$ selection rule. As the majority of the population enters this dark state, the vapor becomes transparent to the laser light [39].

This pumped atomic vapor achieves a highly polarized state. The magnetometer's performance fundamentally depends on maintaining the coherence of this polarized state, which is constantly degraded by various relaxation processes, including spin-exchange collisions, atomic diffusion, and magnetic field inhomogeneities [39].

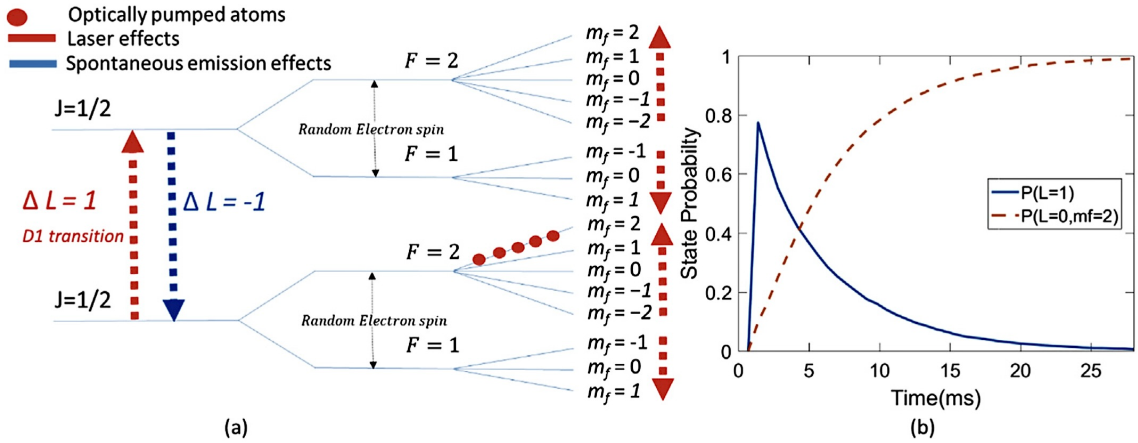


Figure 2.5 (a) Energy level diagram for ^{87}Rb, showing how circularly polarized D1 laser light drives transitions with $\Delta m_F = +1$, while spontaneous emission allows for probabilistic decays ($\Delta m_F = 0, \pm 1$). **(b)** The resulting population dynamics, illustrating how atoms accumulate in the non-absorbing "dark state" ($F = 2, m_F = 2$), rendering the vapor transparent [39, Fig. 3].

To optimize sensitivity, these effects must be managed. This can be done by increasing the spin-exchange rate through higher atomic density or by minimizing the ambient magnetic field. Field minimization is a multi-stage process involving passive shielding (e.g., a magnetically shielded room) to reduce the field to a residual level (typically ~50 nT). Since OPMs operate in a quasi-zero-field mode, this residual field must be nulled. To accomplish this, an integrated set of three-axis DC coils around the vapor cell is actively driven by an automated algorithm. This zeroing procedure is a preparatory step, executed before primary data acquisition begins. The algorithm finds and applies the necessary DC currents to the coils to counteract the static ambient field, ensuring that the sensor operates in a pre-established, near-zero-field environment during subsequent measurements. This calibration is repeated whenever the external field or the sensor's orientation changes significantly, thereby maintaining optimal operating conditions without attenuating the physiological signals of interest [39, 28].

The dynamics of the net magnetization, or polarization $\mathbf{P}$, of this prepared atomic ensemble in an external magnetic field $\mathbf{B}$ are described by the phenomenological Bloch equations.

$$\frac{dP_x}{dt} = \gamma(P_y B_z - P_z B_y) - \frac{P_x}{\tau} \tag{2.4}$$

$$\frac{dP_y}{dt} = \gamma(P_z B_x - P_x B_z) - \frac{P_y}{\tau} \tag{2.5}$$

$$\frac{dP_z}{dt} = \gamma(P_x B_y - P_y B_x) + \frac{P_0 - P_z}{\tau} \tag{2.6}$$

This general model accounts for the precession of the polarization around the magnetic field (at a rate determined by the gyromagnetic ratio γ), the optical pumping process driving the polarization towards an equilibrium value P_0 along the pump axis (here, z), and various relaxation mechanisms (e.g., spin-exchange collisions, diffusion). These competing effects are typically simplified by combining them into a single effective relaxation time, τ. The measured signal is the light transmission, which is proportional to the polarization component along the laser's propagation axis (P_z). Consequently, a high polarization (high transparency) results in a high voltage on the photodiode [39].

Assuming a static, transverse magnetic field ($B_z = 0$, $B_x \neq 0$, $B_y \neq 0$), we substitute these components into the Bloch equations:

$$\frac{dP_x}{dt} = -\gamma P_z B_y - \frac{P_x}{\tau} \tag{2.7}$$
$$\frac{dP_y}{dt} = \gamma P_z B_x - \frac{P_y}{\tau} \tag{2.8}$$
$$\frac{dP_z}{dt} = \gamma(P_x B_y - P_y B_x) + \frac{P_0 - P_z}{\tau} \tag{2.9}$$

To find the steady-state solution, we set all time derivatives to zero ($\frac{d\mathbf{P}}{dt} = 0$). This transforms the system of differential equations into a system of algebraic equations [39]:

$$0 = -\gamma P_z B_y - \frac{P_x}{\tau} \tag{1}$$
$$0 = \gamma P_z B_x - \frac{P_y}{\tau} \tag{2}$$
$$0 = \gamma(P_x B_y - P_y B_x) + \frac{P_0 - P_z}{\tau} \tag{3}$$

Solving this system yields the final expressions for the three components of the polarization:

$$\boxed{P_x = \frac{-\gamma\tau B_y P_0}{1 + (\gamma\tau)^2(B_x^2 + B_y^2)}} \tag{2.10}$$

$$\boxed{P_y = \frac{\gamma\tau B_x P_0}{1 + (\gamma\tau)^2(B_x^2 + B_y^2)}} \tag{2.11}$$

$$\boxed{P_z = \frac{P_o}{1 + (\gamma\tau)^2(B_x^2 + B_y^2)}} \tag{2.12}$$

The polarization along the pump axis, P_z, follows a Lorentzian absorption profile. However, this DC measurement scheme presents two significant drawbacks: it is highly sensitive to low-frequency 1/f noise, and because the photodiode only measures the total absorption (P_z), it cannot distinguish the individual contributions of B_x and B_y. The solution is to use a lock-in detection scheme by producing an amplitude modulation of the polarization [39].

To achieve this, a modulation field is applied along the x-and y-directions. Since only P_z is measured by the photodiode, we focus on the equation for this component:

$$\frac{dP_z}{dt} = \gamma P_x(B_y + B_1 \cdot \sin(\omega_{\mathsf{mod}} t)) - \gamma P_y(B_x + B_2 \cdot \sin(\omega_{\mathsf{mod}} t + \pi/2)) + \frac{P_o - P_z}{\tau} \tag{2.13}$$

While solving the full system is complex, the principle can be understood by examining a simpler case. It can be shown that an approximate solution to the differential equation system:

$$\frac{dP_z}{dt} = \gamma(P_x(B_y + B_1 \cdot \sin(\omega_{\mathsf{mod}} t))) + \frac{P_o - P_z}{\tau} \tag{2.14}$$
$$\frac{dP_x}{dt} = -\gamma P_z B_y - \frac{P_x}{\tau} \tag{2.15}$$

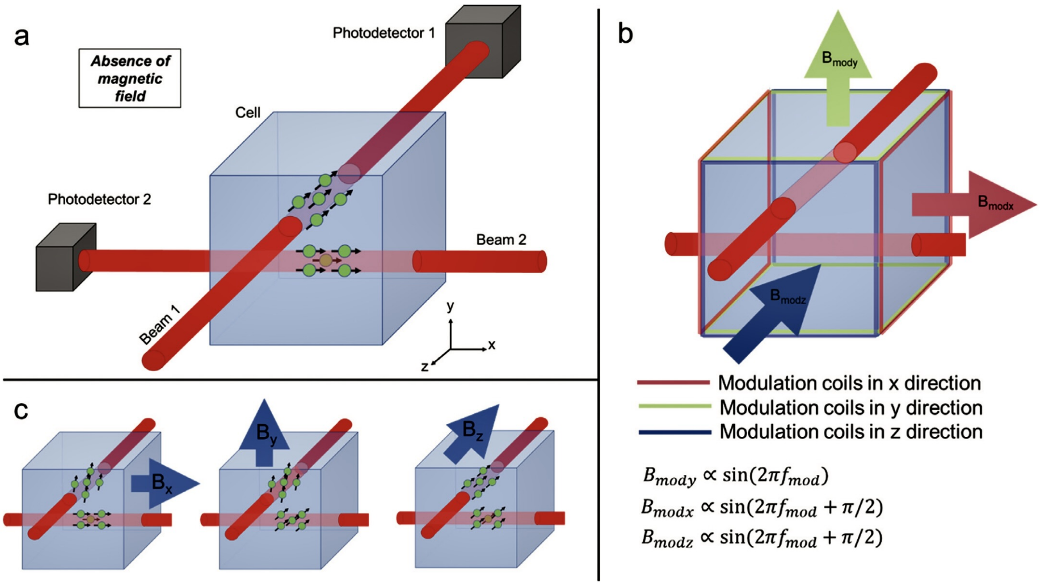


Figure 2.6 (a) Schematic of a dual-axis OPM sensor where a single 795nm laser is split to create two independent, circularly polarized pump beams for the ^{87}Rb vapor cell. **(b)** On-board coils generate orthogonal modulation fields (0° and 90° phase shift) in the x and y directions. **(**c**)** This configuration allows Beam 1 to detect x and y fields, while Beam 2 detects y and z fields, creating a tri-axial measurement capability [3, Fig. 1].

is of the form [39]:

$$P_z(t) = P_0' J_0\left(\frac{\gamma B_1}{\omega}\right) J_1\left(\frac{\gamma B_1}{\omega}\right) \frac{\gamma B_y \tau}{1 + (\gamma B_y \tau)^2} \sin \omega t. \tag{2.16}$$

The full equation for dP_z/dt also contains a term solely dependent on P_y and B_x. Since the equations for both components have essentially the same form and differentiation is a linear operation, the total solution is expected to be a superposition of individual solutions. As the measured voltage is proportional to the polarization in the z-direction, we thus arrive at [3]:

$$V(t) = A_1 \frac{\gamma B_y \tau}{1 + (\gamma B_y \tau)^2} \sin(\omega t) + A_2 \frac{\gamma B_x \tau}{1 + (\gamma B_x \tau)^2} \sin(\omega t + \pi/2) \tag{2.17}$$

where A_1 and A_2 are proportionality constants determined through calibration. The two signal components are phase-shifted by 90° (in quadrature), which allows a lock-in amplifier to differentiate between them [3]. Furthermore, because these signals possess a high frequency, this method achieves a high signal-to-noise ratio by avoiding low-frequency noise. If we now assume the magnetic fields are very small, the Lorentzian terms can be approximated as linear, as shown in Figure 2.7. The demodulated outputs from the lock-in amplifier ultimately become [39]:

$$V_1(t) \approx A_1 \gamma \tau B_y$$
$$V_2(t) \approx A_2 \gamma \tau B_x$$

It should be noted that this linearity holds only when the field components are close to zero; large static fields can introduce non-linearity and calibration errors, necessitating robust active shielding. This linear relationship highlights a fundamental trade-off: increasing sensitivity (the slope, proportional to τ) reduces the linear dynamic range of the sensor. The optimal compromise is application-specific [39].

A single sensor using one optical axis can thus measure two orthogonal magnetic field components. To create a tri-axial sensor, a second, perpendicular pump-probe beam can be added, which would be sensitive to the remaining field components (e.g., B_y and B_z), with the common-axis measurement (B_y) being averaged. This dual-beam configuration is shown in Figure 2.6 [3].

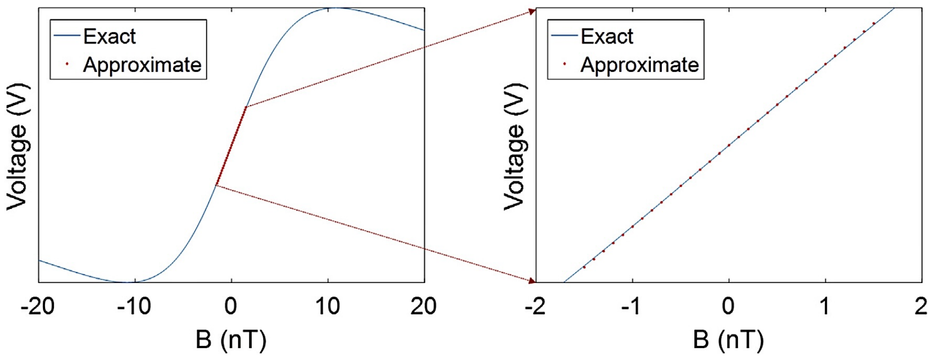


Figure 2.7 System response over varying field strengths. Left: The sensor's response over a 40 nT range shows clear non-linearity. Right: Within the sensor's typical dynamic range (±1.5 nT), a linear approximation remains highly accurate, deviating by less than 1% at 1 nT. This assumes negligible transverse fields [39, Fig. 5].

When multiple sensors operate in parallel in a multi-channel system, the challenge of crosstalk arises from two sources: the AC modulation fields and the DC compensation coils. The AC modulation field from one sensor can be picked up by an adjacent sensor, altering its sensitive axis and calibration. However, these effects are often negligible if sensors are placed more than 2 cm apart. Similarly, the DC compensation field from one sensor can shift a neighbor out of its operational zero-field range. To mitigate this DC crosstalk, a global optimization strategy is employed where the field-zeroing procedure is run simultaneously on all sensors in the array. This collapses the system to a state where each sensor experiences a near-zero field. The downside of this method is its lack of modularity; reconfiguring a single sensor requires repeating the calibration for the entire system [28].

2.3 Introduction to Independent Component Analysis (ICA)

In signal processing, a common task is to resolve mixed signals or remove noise from data. Methods of *Blind Source Separation (BSS)*, particularly *Independent Component Analysis (ICA)*, provide a powerful framework for this task. This statistical process is often explained using the "cocktail party problem" as an analogy: the challenge of isolating one speaker's voice from a cacophony of others in a crowded room.

Consider a scenario where a primary speaker is addressing an audience, but ten individuals within that audience are whispering among themselves. To capture the speaker's voice with high fidelity, eleven microphones are strategically placed in the room. Inevitably, each microphone records a unique blend of all eleven sound sources: the speaker and the ten whispering individuals. The objective of ICA is to take these mixed recordings and computationally separate them, isolating not only the speaker's signal but also the distinct signals of the ten other individuals, using only the microphone measurements.

This scenario can be modeled mathematically. Let the original, unknown source signals (the speaker and the ten whisperers) be represented by a vector $\mathbf{s}(t) = [s_1(t), s_2(t), ..., s_{11}(t)]^T$. The signals captured by the microphones are denoted by another vector, $\mathbf{x}(t) = [x_1(t), x_2(t), ..., x_{11}(t)]^T$. The relationship between the original sources and the measured signals is a linear mixture:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) \tag{2.18}$$

In this equation, $\mathbf{A}$ is an unknown 11×11 "mixing matrix." Each element a_{ij} within this matrix is a physical scaling factor that accounts for variables such as the distance from the j-th person to the i-th microphone, the microphone's gain, and the acoustic properties of the room. The fundamental challenge is that both the mixing matrix $\mathbf{A}$ and the source signals $\mathbf{s}(t)$ are unknown; the only available information is the mixed signals $\mathbf{x}(t)$ [13].

If the mixing parameters were known, one could simply solve the linear system by inverting the matrix $\mathbf{A}$. However, since both $\mathbf{A}$ and $\mathbf{s}(t)$ are unknown, the problem is considerably more complex. The solution lies in leveraging the statistical properties of the signals. The core idea behind ICA is to find an "unmixing" matrix that, when applied to the observed signals, produces components that are as statistically independent as possible.

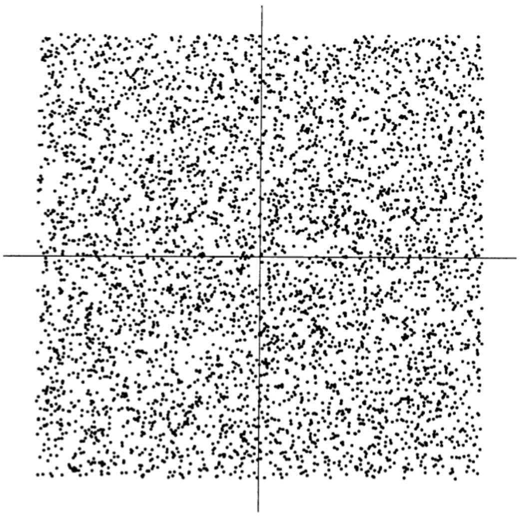
To illustrate the ICA model in statistical terms, consider a simplified case with two independent source signals, s_1 and s_2. We assume they each follow a uniform probability distribution, chosen so that each signal has a mean of zero and a variance of one:

$$p(s_i) = \begin{cases} \frac{1}{2\sqrt{3}} & \text{if } |s_i| \leq \sqrt{3} \\ 0 & \text{otherwise} \end{cases} \tag{2.19}$$

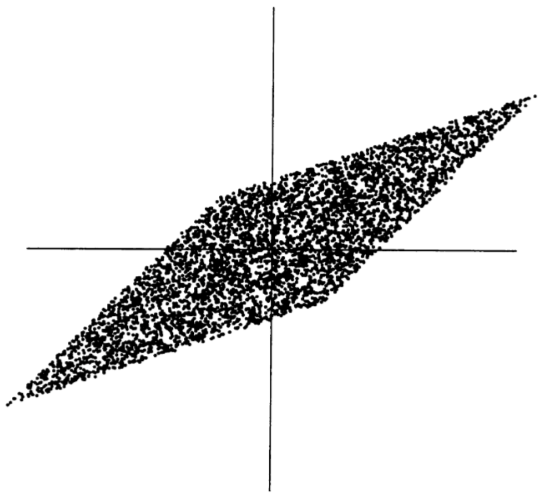
Because s_1 and s_2 are independent, their joint probability density is uniform over a square, as shown in Figure 2.8a. Now, let's mix these two independent components using the following mixing matrix:

$$\mathbf{A}_0 = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \tag{2.20}$$

This linear transformation creates two new mixed variables, x_1 and x_2. The joint distribution of this new, mixed data is now uniform over a *parallelogram* instead of a square. As a result, the random variables x_1 and x_2 are no longer independent. This is intuitively clear: if x_1 attains its maximum or minimum value (a corner of the parallelogram), the value of x_2 is completely determined (see Figure 2.9b). The central problem of ICA is now evident: given only the mixed data (the points forming the parallelogram), we must estimate the original mixing matrix $\mathbf{A}_0$. In this simple case, the edges of the parallelogram are aligned with the directions of the column vectors of the mixing matrix. However, this geometric approach only works for uniform distributions, and a more robust statistical framework is required for general applications [13].



(a)



(b)

Figure 2.8 (a) The joint distribution of the independent components s_1 and s_2 with uniform distributions. Horizontal axis: s_1, vertical axis: s_2. [13, Fig. 5] **(b)** The joint distribution of the observed mixtures x_1 and x_2. Horizontal axis: x_1, vertical axis: x_2 [13, Fig. 6]

For the basic ICA model to be estimable, certain assumptions must be made [13]:

Statistical Independence: The cornerstone of ICA is the assumption that the source signals are statistically independent. This means that knowing the value of one source signal provides no information about the values of the others.

Nongaussian Distributions: The independent source signals must follow non-Gaussian distributions. Gaussian distributions are symmetric and lack the higher-order statistical information necessary for ICA to distinguish between the sources, as will be shown later.

Square and Invertible Mixing Matrix: For simplicity, the basic model assumes that the number of observed mixtures is equal to the number of independent components, resulting in a square mixing matrix that is also assumed to be invertible.

In this fundamental model, any time delays or time dependence in the mixing process are disregarded, which is why it is often referred to as the *instantaneous mixing model*. Consequently, the time index t is frequently dropped, and each mixture x_i and independent component s_j are treated as random variables rather than time signals.

The nature of the ICA problem leads to two inherent ambiguities [13]:

1. **We cannot determine the variances (energies) of the independent components.**

   Because both $\mathbf{s}$ and $\mathbf{A}$ are unknown, any scalar multiplier in one of the sources s_i can be cancelled by dividing the corresponding column $\mathbf{a}_i$ of $\mathbf{A}$ by the same scalar. As a consequence, the magnitudes of the independent components are typically fixed by assuming each has a unit variance: $E\{s_i^2\} = 1$. The matrix $\mathbf{A}$ is then adapted in the solution to account for this restriction. This still leaves a sign ambiguity, as we could multiply an independent component by -1 without affecting the model. This ambiguity is insignificant in most applications.

2. **We cannot determine the order of the independent components.**

   Since $\mathbf{s}$ and $\mathbf{A}$ are unknown, we can freely change the order of the terms in the sum $\mathbf{x} = \mathbf{A}\mathbf{s}$. Formally, a permutation matrix $\mathbf{P}$ and its inverse can be substituted to give $\mathbf{x} = \mathbf{A}\mathbf{P}^{-1}\mathbf{P}\mathbf{s}$. The vector $\mathbf{P}\mathbf{s}$ contains the original independent variables in a new order, and the matrix $\mathbf{A}\mathbf{P}^{-1}$ becomes a new unknown mixing matrix.

Statistical independence is formally defined using probability density functions (PDFs). For two random variables, y_1 and y_2, to be independent, their joint PDF, $p(y_1, y_2)$, must be factorizable into the product of their marginal PDFs:

$$p(y_1, y_2) = p_1(y_1)p_2(y_2) \tag{2.21}$$

This principle extends to any number of random variables. It is crucial to distinguish independence from a weaker condition known as *uncorrelatedness*, where the covariance is zero:

$$E\{y_1 y_2\} - E\{y_1\}E\{y_2\} = 0 \tag{2.22}$$

While independent variables are always uncorrelated, the converse is not necessarily true [13].

A standard preprocessing step in ICA is to first center the data by subtracting its mean and then to "whiten" it. Whitening is a linear transformation that makes the components of the observed vector $\mathbf{x}$ uncorrelated and gives them unit variance. The resulting vector $\tilde{\mathbf{x}}$ satisfies:

$$E\{\tilde{\mathbf{x}}\tilde{\mathbf{x}}^T\} = \mathbf{I} \tag{2.23}$$

One popular method for whitening is to use the eigenvalue decomposition (EVD) of the covariance matrix $E\{\mathbf{x}\mathbf{x}^T\} = \mathbf{E}\mathbf{D}\mathbf{E}^T$ [13]. This is always possible as the covariance matrix is defined to be real and symmetric. Whitening is then performed by:

$$\tilde{\mathbf{x}} = \mathbf{E}\mathbf{D}^{-1/2}\mathbf{E}^T\mathbf{x} \tag{2.24}$$

where $\mathbf{D}^{-1/2}$ is computed by taking the inverse square root of each eigenvalue. The utility of whitening is that it transforms the mixing matrix $\mathbf{A}$ into a new orthogonal matrix $\tilde{\mathbf{A}}$. Since the sources $\mathbf{s}$ are assumed to have unit variance and are independent, their covariance matrix is the identity matrix ($E\{\mathbf{s}\mathbf{s}^T\} = \mathbf{I}$). This leads to:

$$E\{\tilde{\mathbf{x}}\tilde{\mathbf{x}}^T\} = \tilde{\mathbf{A}}E\{\mathbf{s}\mathbf{s}^T\}\tilde{\mathbf{A}}^T = \tilde{\mathbf{A}}\tilde{\mathbf{A}}^T = \mathbf{I} \tag{2.25}$$

This reduces the complexity of the problem significantly. Instead of estimating the n^2 elements of $\mathbf{A}$, we only need to estimate the orthogonal matrix $\tilde{\mathbf{A}}$, which has only $n(n-1)/2$ degrees of freedom. Thus, whitening solves half of the ICA problem. For the remainder of this text, we assume the data has been preprocessed and, for simplicity, use $\mathbf{x}$ and $\mathbf{A}$ to denote the whitened data and the new orthogonal mixing matrix [13].

The necessity of the non-Gaussianity assumption becomes clear in the context of whitened data. Let's assume our independent components, s_1 and s_2, are standard Gaussian variables. Their joint PDF is radially symmetric:

$$p(s_1, s_2) = \frac{1}{2\pi}\exp\left(-\frac{s_1^2 + s_2^2}{2}\right) = \frac{1}{2\pi}\exp\left(-\frac{\|\mathbf{s}\|^2}{2}\right) \tag{2.26}$$

After mixing with an orthogonal matrix $\mathbf{A}$ (the case for whitened data), the density of the mixtures $\mathbf{x}$ is:

$$p(x_1, x_2) = \frac{1}{2\pi}\exp\left(-\frac{\|\mathbf{A}^T\mathbf{x}\|^2}{2}\right)|\det \mathbf{A}^T| \tag{2.27}$$

Due to the properties of an orthogonal matrix, $\|\mathbf{A}^T\mathbf{x}\|^2 = \|\mathbf{x}\|^2$ and $|\det \mathbf{A}^T| = 1$. This simplifies the equation to:

$$p(x_1, x_2) = \frac{1}{2\pi}\exp\left(-\frac{\|\mathbf{x}\|^2}{2}\right) \tag{2.28}$$

The probability distribution of the mixed signals is identical to the distribution of the original sources, and the mixing matrix $\mathbf{A}$ has disappeared from the expression. Therefore, there is no way to infer the mixing matrix from the observed mixtures if the sources are Gaussian [12].

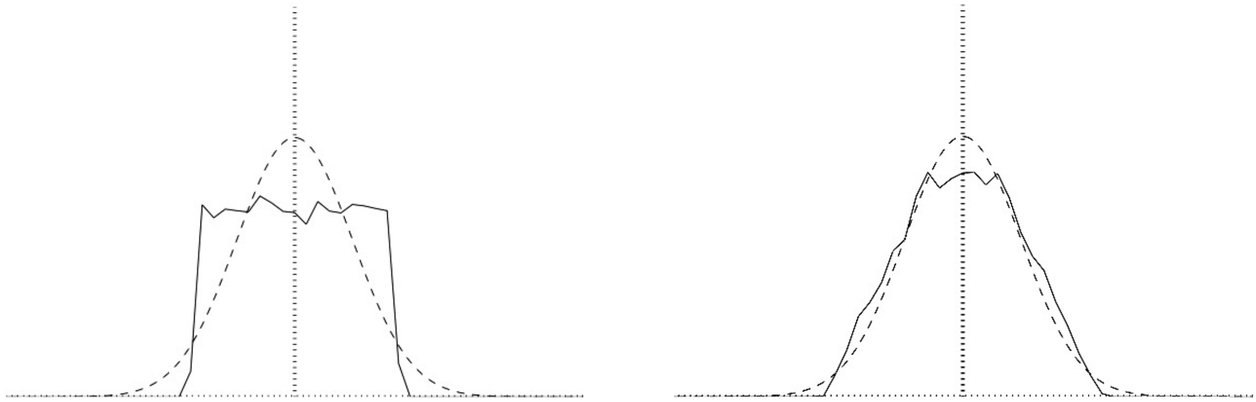
Once the data is preprocessed, the core task of ICA is to estimate the unmixing matrix, which we will call. $\mathbf{W} = \mathbf{A}^T$. This can be accomplished through several distinct theoretical frameworks. The principal strategies include maximizing the non-Gaussianity of the estimated components, minimizing their mutual information, or employing maximum likelihood estimation. While each approach yields a slightly different solution for the unmixing matrix, they all share the common goal of finding a linear transformation that recovers the original sources [13]. This chapter will focus on the first and most intuitive approach: maximizing nongaussianity.

The estimation strategy leverages the Central Limit Theorem (CLT), which states that a sum of independent random variables tends toward a Gaussian distribution. This is exemplified in Figures 2.9a and 2.9 using the previous example of uniform source distributions. ICA operates on the reverse principle: *a mixture of signals is "more Gaussian" than any of the original source signals*. The sources are found by seeking projections of the mixed data that are maximally non-Gaussian. To estimate a single source, we define a linear projection $y = \mathbf{w}^T\mathbf{x} = \mathbf{w}^T\mathbf{A}\mathbf{s}$. This output y is a linear combination of the original sources. The distribution of y becomes *least* Gaussian when it is no longer a sum—that is, when y becomes proportional to a single independent component. The problem thus transforms into an optimization task: find the projection vector $\mathbf{w}$ that maximizes the non-Gaussianity of $\mathbf{w}^T\mathbf{x}$ and, in turn, is a column of the orthogonal mixing matrix $\mathbf{A}$ (i.e., a row of the unmixing matrix $\mathbf{W}$) [13].

A classical metric for quantifying the non-Gaussianity of a probability distribution is its fourth-order cumulant, known as *kurtosis*. For a random variable y, kurtosis is formally defined as [13]:

$$\text{kurt}(y) = \mathbb{E}\{y^4\} - 3(\mathbb{E}\{y^2\})^2 \tag{2.29}$$

When the variable y has been standardized to have zero mean and unit variance ($\mathbb{E}\{y^2\} = 1$), this expression simplifies to kurt$(y) = \mathbb{E}\{y^4\} - 3$. By this definition, a Gaussian distribution has zero kurtosis.



(a) **(b)**

Figure 2.9 (a) Estimated density of a single uniform independent component, shown alongside a Gaussian density (dashed curve) for comparison. (See Figure 2.8a for the corresponding joint distribution.) **(b)** Marginal density of one component from the whitened mixture. After whitening, the joint distribution is expected to resemble a rotated square rather than a parallelogram (See Figure 2.9b). The marginal is closer to the Gaussian density (dashed curve) than the marginals of the original independent components.

Distributions with positive kurtosis are described as *super-Gaussian* (or leptokurtic), often characterized by 'heavy tails' and a sharp peak, while those with negative kurtosis are termed *sub-Gaussian* (or platykurtic), typically appearing flatter and more 'box-shaped' than a Gaussian. In practice, the absolute value of the kurtosis is often used as the measure of nongaussianity. However, a significant limitation of kurtosis is its lack of robustness, as its value can be heavily skewed by outliers in the data [13].

A more sophisticated and robust measure of nongaussianity is *negentropy*, which is rooted in information theory. The foundation of negentropy lies in the concept of differential entropy, defined for a random variable y with a probability density function $f(y)$ as [13]:

$$H(y) = -\int f(y) \log f(y) dy \tag{2.30}$$

A fundamental principle of information theory is that among all random variables with the same variance, the Gaussian variable has the largest entropy. Negentropy, denoted $J(y)$, capitalizes on this by measuring the 'entropy deficit' of a given distribution compared to a Gaussian one. It is defined as the difference between the entropy of a Gaussian variable, y_{gauss}, with the same variance and the entropy of the variable y itself:

$$J(y) = H(y_{\text{gauss}}) - H(y) \tag{2.31}$$

By definition, negentropy is always non-negative and is zero only when y is Gaussian. Since calculating negentropy directly is computationally demanding, it is often estimated using approximations. These approximations typically take the form [13]:

$$J(y) \propto [\mathbb{E}\{G(y)\} - \mathbb{E}\{G(\nu)\}]^2 \tag{2.32}$$

where ν is a standardized Gaussian variable and G is a non-quadratic function. The selection of G is crucial for ensuring robustness against outliers; functions that exhibit slower growth, such as $G_1(u) = \frac{1}{a_1} \log \cosh(a_1 u)$ and $G_2(u) = -\exp(-u^2/2)$, are preferred because they are less sensitive to extreme values [13].

Thus the task is to find a vector $\mathbf{w}$ that maximizes the negentropy approximation $J(\mathbf{w}^T\mathbf{x})$ subject to the constraint that $\mathbf{w}$ has unit norm, i.e., $||\mathbf{w}||^2 = 1$ (both $\mathbf{A}$ and $\mathbf{W} = \mathbf{A}^T$ are orthogonal). This is a constrained optimization problem solvable with Lagrange multipliers.

For an objective function $f(x)$ and a constraint $g(x) = 0$, any solution x_0 must satisfy $\nabla f(x_0)+\lambda_0 \nabla g(x_0) = 0$ for some scalar λ_0 (Theorem of Lagrange-Multipliers). In our case, the objective function is $J(\mathbf{w}^T\mathbf{x})$ (2.29) and the constraint is $g(\mathbf{w}) = ||\mathbf{w}||^2 - 1 = 0$. The gradients are [12]:

$$\nabla_{\mathbf{w}} J(\mathbf{w}^T\mathbf{x}) \propto \mathbb{E}[G'(\mathbf{w}^T\mathbf{x}) \cdot \mathbf{x}] \tag{2.33}$$

$$\nabla_{\mathbf{w}}(\|\mathbf{w}\|^2 - 1) = 2\frac{\mathbf{w}}{||\mathbf{w}||} \tag{2.34}$$

The equation system is then of the following form (with the proportionality absorbed into the Lagrange multiplier λ):

$$\begin{cases} \mathbb{E}\left[G'\left(\boldsymbol{w}^\top \boldsymbol{x}\right) \cdot \boldsymbol{x}\right] + 2\lambda \dfrac{\boldsymbol{w}}{\|\boldsymbol{w}\|} = 0 & \textbf{(I)} \\ \|\boldsymbol{w}\| = 1 & \textbf{(II)} \end{cases}$$

We can now define $\beta = 2 \cdot \lambda$ and insert equation **II** into **I**. We thus arrive at [12]:

$$\mathbb{E}\left[G'(\boldsymbol{w}^\top \boldsymbol{x}) \cdot \boldsymbol{x}\right] + \beta \cdot \boldsymbol{w} = 0 \tag{2.35}$$

Solving for the optimal $\mathbf{w}$ can be done numerically using a Newton-like method. After some simplification and approximation, we arrive at the following update rule [12]:

$$\mathbf{w}^+ \leftarrow \mathbb{E}\left\{\mathbf{x}G'(\mathbf{w}^T\mathbf{x})\right\} - \mathbb{E}\left\{G''(\mathbf{w}^T\mathbf{x})\right\}\mathbf{w} \tag{2.36}$$

The FastICA algorithm for finding a single component can be summarized as (Note that we have to renormalize after each step as the update rule only holds approximately) [13]:

1. Choose an initial (e.g., random) weight vector $\mathbf{w}$.
2. Let $\mathbf{w}^+ = \mathbb{E}\left\{\mathbf{x}g(\mathbf{w}^{\mathrm{T}}\mathbf{x})\right\} - \mathbb{E}\left\{g'(\mathbf{w}^{\mathrm{T}}\mathbf{x})\right\}\mathbf{w}$.
3. Let $\mathbf{w} = \frac{\mathbf{w}^+}{\|\mathbf{w}^+\|}$.
4. If not converged, go back to step 2.

Convergence occurs when the old and new values of $\mathbf{w}$ point in the same direction (their dot product is close to 1).

To estimate multiple independent components, we can apply this algorithm sequentially. To prevent different vectors $\mathbf{w}_1, \ldots, \mathbf{w}_n$ from converging to the same maximum, we must ensure they are decorrelated after every iteration. For whitened data, this is equivalent to orthogonalization. A simple method is a deflation scheme based on Gram-Schmidt orthogonalization. When estimating the $(p+1)$-th vector, $\mathbf{w}_{p+1}$, its projection onto the previously found vectors $\mathbf{w}_1, \ldots, \mathbf{w}_p$ is subtracted after each update step [13].

Finally, it is crucial to note that ICA is an algorithm fundamentally designed to disentangle scalar time series data. Suppose we want to apply this process component-wise to a vector-valued signal. Due to the inherent permutation ambiguity, the original order of the components is lost. Consequently, identifying specific source signals of corresponding components in the vector signal from the ICA output is only possible if prior knowledge about the sources' characteristics is available. Furthermore, because ICA can only determine each component up to a scalar constant (the magnitude ambiguity), information regarding the absolute amplitude and sign of the original source signals is lost for each component, which leads to a loss of critical information, such as the vectorial direction and magnitude. Applying ICA component-wise to vector-valued signals must therefore be approached with great caution.

To overcome these limitations, we introduce a novel framework that integrates ICA with physiology-informed evaluation. Our method employs a composite score to assess each resulting ICA component, combining a neural network's confidence in heartbeat detection with a plausibility score based on physiological segment durations. This allows us to systematically identify and retain components that are verifiably cardiac-related while discarding those associated with artifacts. By reconstructing the signal using only the retained components and the ICA mixing matrix, we effectively suppress noise and preserve the essential physiological dynamics of the MCG data, as described in Chapter 4.1.

3 Training a deep neural network for MCG segmentation

As outlined in Chapter 1, the electrocardiogram (ECG) is a common method for monitoring the heart's activity. It is characterized by three major segments: the P-wave, QRS complex, and T-wave (Figure 3.1). This segmentation, or *delineation*, is typically performed manually by a cardiologist, but the process is tedious and time-consuming. Consequently, automated methods using wavelet-based algorithms and deep learning have become an active area of research.

This thesis, however, focuses on the magnetocardiography (MCG) signal. While ECG and MCG signals share essential features, the specific analysis herein requires examining the vector magnetocardiogram (VMCG) within very short, averaged heartbeat cycles of approximately 1–2 seconds (Figure 4.4). During this work, the windowing process proved challenging, as naive peak detection methods were easily disrupted by the high noise levels characteristic of MCG recordings.

Although machine learning solutions for ECG delineation exist, they have several limitations for the specific application addressed in this thesis. State-of-the-art solutions are often trained on longer data periods (5–10 seconds), making them less suitable for short-window analysis. Furthermore, many existing models fail to normalize the input data, which is a critical oversight, as MCG and ECG signals have vastly different absolute scales. Finally, current solutions often process several leads simultaneously. We require a model that treats each channel independently because the three directional components of an MCG do not directly correspond to a standard ECG lead.

To address these challenges, this chapter proposes a U-Net-based neural network capable of robustly segmenting both MCG and ECG data. The model is specifically designed to be effective on very short segments (1–2 seconds), to operate reliably on noisy data, and to process signals on a single-channel basis. Beyond segmentation, this chapter also demonstrates how the model enables robust peak detection and a novel signal quality metric, the "Heartbeat Score," which are utilized in subsequent chapters.

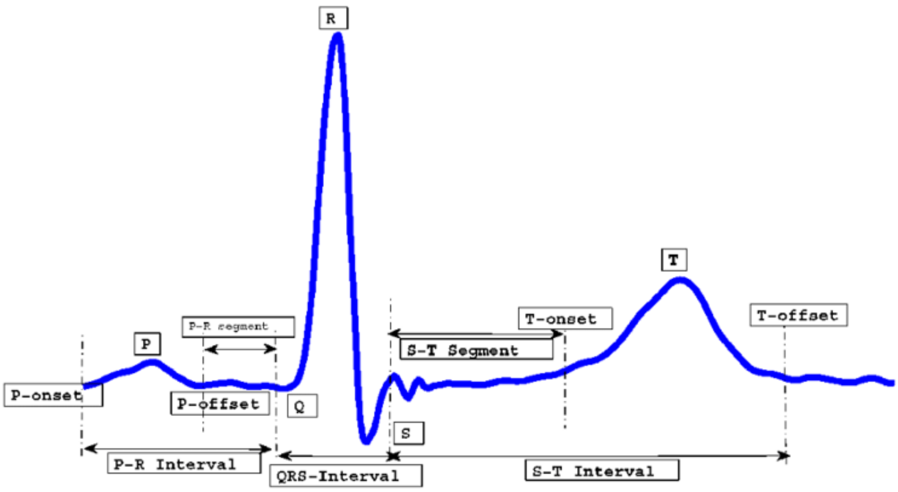


Figure 3.1 An electrocardiogram (ECG) waveform from a single heartbeat, illustrating the P-wave, QRS complex, and T-wave. The onset and offset points for each segment are marked. [18, Fig. 1]

3.1 Datasets and Model Architecture

Datasets and Preprocessing for Model Training

Since no open-source MCG datasets are currently available, and due to the inherent similarity between ECG and MCG data, established ECG datasets were used for training and benchmarking segmentation performance. The two most commonly used open-source ECG databases with corresponding wave delineations are the QT Database (QTDB) [7, 20] and the Lobachevsky University Database (LUDB) [7, 17, 16].

The QT Database (QTDB) [20] is an open-source database first published in 1999, designed for developing and evaluating algorithms that detect waveform boundaries. It consists of 105 15-minute, two-lead ECG recordings sampled at 250 Hz. For each record, at least 30 individual heartbeat cycles were annotated (approximately 30 seconds of data), resulting in over 3,622 annotated beats. The annotations were provided by expert annotators. The database was specifically designed to capture a broad variety of QRS and ST-T morphologies. Some of the recordings were previously published in the MIT-BIH Arrhythmia Database and the European ST-T Database.

The Lobachevsky University Database (LUDB) [17, 16] was released more recently in 2021. It consists of 10-second, 12-lead ECG recordings from 200 healthy volunteers and patients, sampled at 500 Hz and collected by the Nizhny Novgorod City Hospital. It was specifically designed to address several shortcomings of the QTDB. While the QTDB contains annotations for P-, QRS-, and T-waves, many complexes are unmarked, some recordings only have QRS segmentations, and most are missing T-onset information. The LUDB aims to provide more complete annotations and capture a greater variety of morphologies.

A comparison of the key characteristics of both datasets is provided in Table 3.1.

Table 3.1 Key characteristics of the ECG datasets used for model training.

Data Source	# Recordings	Annotated Duration	Frequency	Leads	Annotations
QTDB	105	$\sim$30 seconds	250 Hz	2	P on/offsets, QRS on/offsets, T offsets
LUDB	200	10 seconds	500 Hz	12	P on/offsets, QRS on/offsets, T on/offsets

Data Preprocessing and Preparation

Before training, the data from both datasets required several preprocessing steps. First, to harmonize the data, the LUDB signals were downsampled from 500 Hz to 250 Hz to match the sampling frequency of the QTDB. Each lead from both datasets was then processed independently and treated as a distinct signal.

For the QTDB, records that only contained annotations for the QRS complex were discarded. A more significant issue was the frequent absence of T-wave onset information. To address this, an artificial T-onset was generated for each beat by sampling from a Gaussian distribution centered around the physiologically normal ST segment length of 100 ms [33], with a standard deviation of 20 ms. For the LUDB, which often lacked annotations in the first and last two seconds of a recording, only the annotated intervals were used [15, 26].

The datasets were partitioned into training, validation, and test sets using a strict patient-level split to prevent data leakage across sets. The test set was composed exclusively of records from 40% of the LUDB patients. The validation set, used to guide model selection, was formed from 5% of patients from the QTDB and 5% of the remaining LUDB patients. Consequently, the final training set consisted of all other records, comprising 95% of suitable QTDB patients and the remaining 55% of LUDB patients.

Segmentation Model Architectures

As is commonly proposed in the literature, a U-Net-like architecture was chosen [15, 26]. This is shown in Figure 3.2. The model is designed to process each ECG lead individually, accepting an input tensor of shape $(B, 1, T)$, where B is the batch size and T is the number of time steps. The output is a tensor of shape $(B, 4, T)$, providing a probability distribution over the four considered classes (No Wave, P-Wave, QRS complex, and T-Wave) for each time step. The final segmentation is determined by selecting the class with the maximum predicted probability for each time step.

The architecture consists of a contracting path (encoder), a bottleneck with an attention mechanism, and an expansive path (decoder), which are detailed below.

Encoder: The encoder forms the contracting path of the network. It is composed of a series of convolutional blocks. Each block applies a 1D convolution, followed by batch normalization and a ReLU activation function. After each block, a max-pooling operation is used to downsample the temporal resolution by half, while the number of feature channels is increased. This process allows the model to learn hierarchical features at different scales.

Bottleneck with Multi-Head Self-Attention: At the lowest point of the U-Net, the bottleneck processes the most abstract and compressed feature representation. A key feature of the architecture is the integration of a Multi-Head Self-Attention (MHSA) mechanism within this bottleneck. Following a standard convolutional block, the MHSA layer allows the model to weigh the importance of different features across the entire sequence, capturing global dependencies and long-range contextual information that are crucial for accurately identifying wave boundaries [40].

Decoder and Skip Connections: The decoder, or expansive path, symmetrically mirrors the encoder. It progressively upsamples the feature maps using 1D transposed convolutions, halving the number of channels and doubling the temporal resolution at each step. Crucially, after each upsampling, the feature map is concatenated with the corresponding high-resolution feature map from the encoder via a skip connection. These connections provide the decoder with essential high-resolution spatial information that is often lost during downsampling, enabling precise localization of the cardiac waves in the final output. Additionally, it allows the gradients to take a shorter route to the parameters located deep within the encoder layers, making these efficiently tunable.

Output Layer: Finally, a 1x1 convolutional layer is applied to map the feature channels from the last decoder block to the final number of classes (four). The tensor dimensions are then permuted to produce the final output shape of (B, T, C), which is well-suited for a standard cross-entropy loss calculation at each time step.

Model Configurations. To investigate the trade-off between performance and computational complexity, two distinct versions of the model were trained:

- **A larger model (UNet-1D-15M)** designed for higher capacity, featuring four encoder/decoder levels with channel sizes of [64, 128, 256, 512] and approximately 15M parameters. The MHSA mechanism in the bottleneck uses 8 attention heads, and a dropout rate of 0.6 is applied for regularization.
- **A lightweight model (UNet-1D-900k)** for faster inference, configured with three levels, channel sizes of [32, 64, 128], and approximately 900k parameters. It employs a simpler 4-head attention mechanism and uses a dropout rate of 0.5.

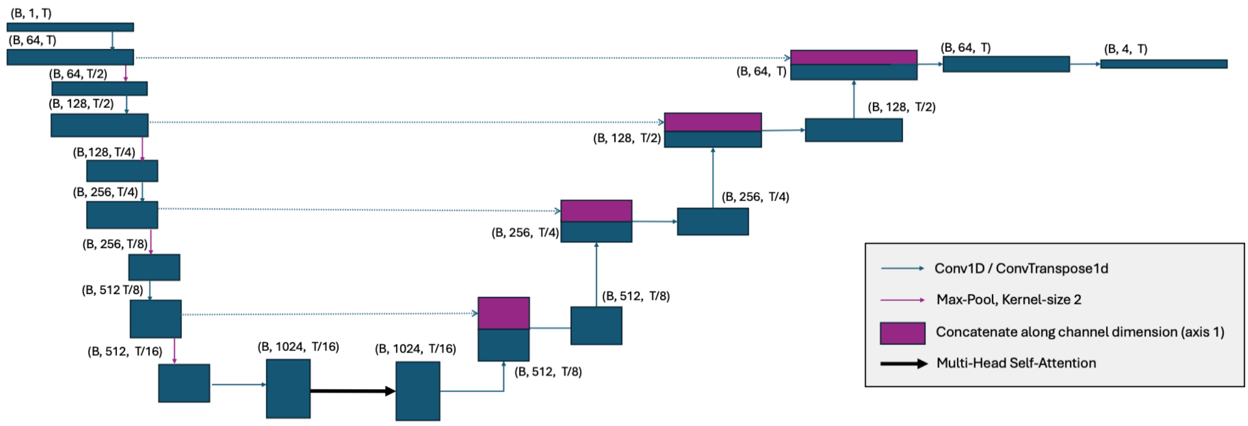


Figure 3.2 Architecture of the U-Net-1D-15M model. The diagram illustrates the data flow and tensor shape transformations, where B is the batch size (e.g., 64) and T is the number of time steps (500, corresponding to a 2-second segment). The dimensions of the blocks represent the number of channels and the temporal length, though they are not drawn to scale.

Postprocessing

As proposed by Joung et al. [15], a simple postprocessing step is employed to remove small, erroneously predicted artifacts rather than using the raw network output directly. It consists of the following steps:

First, contiguous intervals are extracted for each class (P-wave, QRS, T-wave, and no wave) where that class has the highest predicted probability. Next, segments shorter than a 40 ms threshold (approximately one-third of a typical QRS complex [33]) are re-evaluated based on their neighbors using the following rules:

If the two intervals adjacent to a short region have the same label, the short region is assigned that same label, and the three regions are merged into a single segment. If the adjacent intervals have different labels, the short region is assigned the "no wave" label.

From the resulting segments, the corresponding onsets and offsets are determined by selecting the first and last indices of the intervals. Note that no averaging of predictions from multiple leads of the same patient is employed. This would interfere with the ultimate goal of deploying this algorithm to MCG data. Therefore, during evaluation, each lead is treated as a completely independent signal [15].

3.2 Training Procedure

Training Configuration

Both models were trained for 100 epochs using a single NVIDIA Quadro P4000 GPU, with total training times of approximately 1 to 2 hours with a batch size of 64. To select the optimal model and prevent overfitting, a strategy of early stopping was employed. The selection criterion was the sample-wise F1-score on the validation set, which evaluates the classification performance for each individual time step in the signal. The model weights from the epoch yielding the highest sample-wise F_1-score were then retained for the final, event-based evaluation.

Data Loading and Augmentation

For training, all leads in the training dataset were treated as independent signals. They were subdivided into intervals of two-second length, as the objective was to train a model on very short segments. These intervals were created with an overlap of 1.6 seconds, a strategy that dramatically increased the number of available training samples. Notably, no preliminary filtering, such as low-pass, band-pass, or Gaussian filters, was applied. This resulted in approximately 33,000 samples in the training dataset. During testing and validation, however, non-overlapping 5-second intervals were created to ensure compatibility with benchmarks in the literature.

To enhance model generalization and robustness, a series of data augmentations was applied to each 1D signal during training. These augmentations were applied independently, each with a probability of 80% during every training iteration.

Amplitude Scaling: The signal's amplitude was randomly scaled by a factor α, drawn from a predefined uniform distribution $\mathcal{U}(a_{\min}, a_{\max})$:

$$x_{\text{scaled}}[t] = \alpha \cdot x[t].$$

Time Shifting: The input signal $x = [x_0, x_1, \ldots, x_{T-1}]$ was cyclically shifted by a random integer offset s, ensuring no data points are lost:

$$x_{\text{shifted}}[t] = x[(t - s) \pmod{T}].$$

Baseline Wander: To simulate low-frequency physiological drift, a noise signal composed of 1 to 2 low-frequency sinusoids was added:

$$x_{\text{bw}}[t] = x[t] + \sum_{k=1}^{K} A_k \sin(2\pi f_k t + \phi_k), \quad K \in \{1, 2\},$$

where amplitude $A_k \sim \mathcal{U}(-A_{\max}, A_{\max})$, frequency $f_k \sim \mathcal{U}(0, f_{\max})$, and phase $\phi_k \sim \mathcal{U}(0, 2\pi)$.

Gaussian Noise: To model general sensor and electronic noise, zero-mean Gaussian noise was added:

$$x_{\text{gauss}}[t] = x[t] + \epsilon_t, \quad \text{where} \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2).$$

High-Frequency Sinusoidal Noise: To simulate electrical or muscular interference (e.g., from power lines or EMG artifacts), a high-frequency noise signal composed of 2 to 4 sinusoids was added:

$$x_{\text{hf}}[t] = x[t] + \sum_{k=1}^{K} A_k \sin(2\pi f_k t + \phi_k), \quad K \in \{2, 3, 4\},$$

where amplitude $A_k \sim \mathcal{U}(0, A_{\max})$, frequency f_k is drawn from $\{1, 2, \ldots, 29\}$ Hz, and phase $\phi_k \sim \mathcal{U}(0, 2\pi)$.

Signal Normalization: Following the augmentation pipeline, a two-step normalization process was applied. This step is critical because the model is ultimately intended for MCG data. Since the absolute scales of ECG and MCG signals differ vastly, this normalization prevents the model from learning to identify features, such as the QRS complex, based on their absolute amplitude:

1. **Zero-Mean Normalization:** The signal is centered by subtracting its mean:

$$x_{\text{zero}}[t] = x_{\text{aug}}[t] - \frac{1}{T} \sum_{i=0}^{T-1} x_{\text{aug}}[i].$$

2. **Max-Absolute Scaling:** The zero-mean signal is scaled to lie within $[-1, 1]$:

$$x_{\text{norm}}[t] = \frac{x_{\text{zero}}[t]}{\max_i |x_{\text{zero}}[i]| + \varepsilon},$$

where x_{aug} is the augmented signal and ε is a small constant (e.g., 10^{-6}) to prevent division by zero.

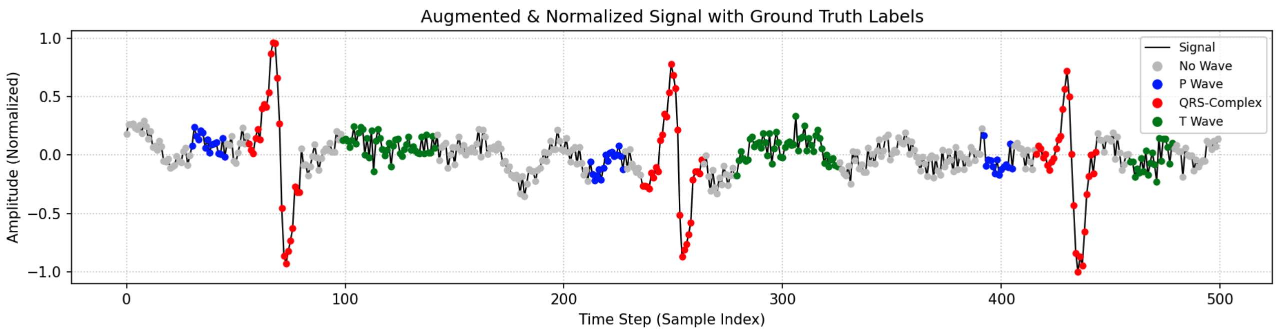


Figure 3.3 Example of the data augmentation pipeline applied to a sample from the training dataset. The original signal is transformed by adding noise, baseline wander, and other augmentations before being normalized.

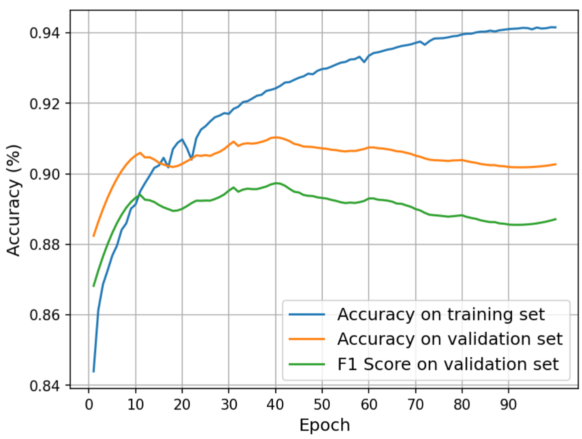
Model Training Configuration

The models were optimized using the *AdamW* optimizer with a weight decay of 10^{-4} to prevent overfitting [19]. A *cosine annealing learning rate schedule* was adopted, starting from a maximum learning rate of 10^{-3} and gradually reducing to a minimum of 10^{-5} over the course of training, as shown in Figure 3.4b.

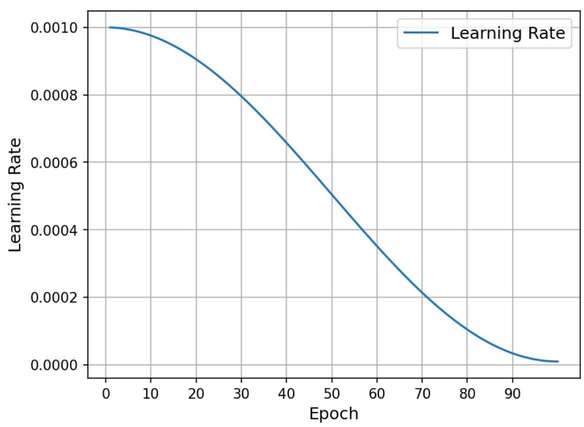
The loss function used was the *Focal Loss*, which is particularly well-suited for handling the class imbalance common in cardiac signal datasets, where "No Wave" segments often dominate. The focal loss is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

where p_t is the predicted probability for the target class, α_t is a weighting factor, and γ is the focusing parameter that adjusts the rate at which easy examples are down-weighted [22].



(a)



(b)

Figure 3.4 (a) Training and smoothed validation accuracy curves over 100 epochs. As shown, the peak validation performance is reached around epoch 40. **(b)** The cosine annealing learning rate schedule, decaying from 10^{-3} to 10^{-5}.

3.3 Evaluation of Model Performance

The model's performance was evaluated on non-overlapping 5-second segments from the test dataset, without the use of data augmentation. An event-based approach was chosen to provide a more nuanced assessment than sample-wise classification accuracy. This approach, which is standard in the literature for cardiac signal analysis, focuses on the model's ability to precisely locate the onsets and offsets of clinically significant events.

The evaluation procedure adheres to the standardized protocol recommended by the Association for the Advancement of Medical Instrumentation (AAMI) [15, 26, 35]. According to this standard, a significant point, defined as the onset or offset of a P, QRS, or T wave, is considered correctly detected if it is located within a ± 150 ms tolerance window of the corresponding expert annotation. Based on this criterion, detection outcomes are categorized as follows:

- **True Positive (TP):** An algorithm-detected point that falls within the ± 150 ms tolerance window of a ground-truth annotation.
- **False Positive (FP):** An algorithm-detected point for which no ground-truth annotation exists within its ± 150 ms window (Type I error).
- **False Negative (FN):** A ground-truth annotation that the algorithm fails to detect within the tolerance window (Type II error).

In line with established conventions in the field [15, 26, 35, 17], several quality metrics are reported. These include the mean error (m) and its standard deviation (σ), calculated from the time deviations for all true positive detections. Additionally, the primary metrics for performance are Sensitivity (Se), Positive Predictive Value (PPV), and the F1-Score.

- **Sensitivity (Se)**, or Recall, measures the proportion of all annotated events that were correctly detected:

$$\text{Se} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.1}$$

- **Positive Predictive Value (PPV)**, or Precision, measures the proportion of the model's detections that were correct:

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.2}$$

- **F1-Score** is the harmonic mean of PPV and Se, providing a single score that balances both metrics:

$$\text{F1-Score} = 2 \cdot \frac{\text{PPV} \cdot \text{Se}}{\text{PPV} + \text{Se}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{3.3}$$

The performance of both proposed models was evaluated on the test dataset. Table 3.2 presents these results, benchmarking them against several state-of-the-art algorithms from the literature. The competing methods primarily employ deep learning techniques, with the notable exception of the classical wavelet-based approach by Kalyakulina et al. [17].

A direct, one-to-one comparison of the results presented in Table 3.2 is challenging due to significant variations in the experimental setups across the cited works. Key differences include:

- **Training and Test Data:** The datasets used for training and testing differ substantially. For instance, Joung et al. [15] utilized a large private dataset, while Moskalenko et al. [26] used a private extension of the LUDB. Sereda et al. [35] trained on 134 records from LUDB and tested on 66, whereas this study used a combination of the QTDB and a portion of LUDB for training, and tested on approximately 80 records from LUDB.

Table 3.2 Delineation performance of the proposed models and other state-of-the-art algorithms on the LUDB test set. Performance is measured by Sensitivity (Se), Positive Predictive Value (PPV), F1-Score, and the mean error with standard deviation ($m \pm \sigma$).

		P onset	P offset	QRS onset	QRS offset	T onset	T offset
Kalyakulina et al. [17]	Se (%)	98.46	98.46	99.61	99.61	–	98.03
	PPV (%)	96.41	96.41	99.87	99.87	–	98.84
	F1 (%)	97.42	97.42	99.74	99.74	–	98.43
	$m \pm \sigma$ (ms)	-2.7 ± 10.2	0.4 ± 11.4	-8.1 ± 7.7	3.8 ± 8.8	–	5.7 ± 15.5
Sereda et al. [35]	Se (%)	95.20	95.39	99.51	99.50	97.95	97.56
	PPV (%)	82.66	82.59	98.17	97.96	94.81	94.96
	F1 (%)	88.49	88.53	98.84	98.72	96.35	96.24
	$m \pm \sigma$ (ms)	2.7 ± 21.9	-7.4 ± 28.6	2.6 ± 12.4	-1.7 ± 14.1	8.4 ± 28.2	-3.1 ± 28.2
Moskalenko et al. [26] (single channel)	Se (%)	97.38	97.36	99.96	99.96	99.43	99.48
	PPV (%)	95.53	95.52	99.84	99.84	98.88	98.94
	F1 (%)	96.47	96.43	99.90	99.90	99.15	99.21
	$m \pm \sigma$ (ms)	0.9 ± 14.1	-3.5 ± 15.7	2.1 ± 9.8	1.6 ± 9.8	1.3 ± 20.9	-0.3 ± 22.9
Joung et al. [15]	Se (%)	98.16	98.20	99.67	99.97	99.82	99.63
	PPV (%)	96.39	96.36	99.29	99.59	99.66	99.42
	F1 (%)	97.24	97.27	99.48	99.78	99.74	99.52
	$m \pm \sigma$ (ms)	7.4 ± 14.1	-1.8 ± 9.9	6.1 ± 10.5	2.0 ± 10.7	3 ± 25.2	4.5 ± 24.4
This work U-Net-1D-900k	Se (%)	95.58	95.60	99.37	99.50	97.84	97.67
	PPV (%)	94.88	94.90	99.70	99.83	98.33	98.15
	F1 (%)	95.23	95.25	99.53	99.67	98.08	97.91
	$m \pm \sigma$ (ms)	2.5 ± 17.3	1.3 ± 16.5	0.2 ± 7.4	-1.2 ± 10.7	-0.3 ± 26.5	-0.8 ± 24.2
This work U-Net-1D-15M	Se (%)	96.12	96.12	99.30	99.43	98.91	98.72
	PPV (%)	96.10	96.10	99.55	99.68	98.98	98.79
	F1 (%)	96.11	96.11	99.42	99.56	98.95	98.75
	$m \pm \sigma$ (ms)	-0.2 ± 17.2	2.5 ± 16.3	1.0 ± 7.8	-1.5 ± 10.6	-0.5 ± 26.1	3.2 ± 23.0

- **Input Signal Segments:** The length of signal intervals varies. The proposed models were trained on very short 2-second segments and evaluated on 5-second segments. In contrast, other works like Joung et al. [15] and Moskalenko et al. [26] used 10-second and 5-second intervals, respectively. Small-scale experiments in this work suggested that performance increases with testing interval length up to 6 seconds.
- **Methodology:** Some approaches employ different signal processing techniques. Kalyakulina et al. [17], for example, utilized all 12 ECG channels and averaged predictions to mitigate errors, a technique not employed in this single-channel approach.

Analysis of Model Performance

Despite these factors, the proposed architecture achieves performance comparable to state-of-the-art solutions, even when trained on a less diverse dataset and with significantly shorter training intervals. This is most likely due to the introduction of the Multi-Head Self-Attention mechanism, which is capable of managing long-range dependencies in the signal.

A notable weakness, however, is observed in P-wave delineation. Compared to state-of-the-art results, both proposed models show a lower F_1-Score, indicating a higher rate of false positive and false negative detections. This performance gap is likely a direct consequence of the training data composition. The inclusion of the QTDB, where nearly every annotated heartbeat contains a P-wave, appears to have induced a model bias. This leads the model to over-predict P-waves when evaluated on the more diverse LUDB test set, which includes arrhythmic recordings where P-waves may be absent. This inclusion was a deliberate trade-off. Leveraging the QTDB was found to be crucial for achieving robust performance on the primary target data (magnetocardiography signals), as a subset of the LUDB recordings exhibits high-frequency and atypically sharp artifacts, as shown in Figure 3.5, not representative of typical ECG/MCG signals.

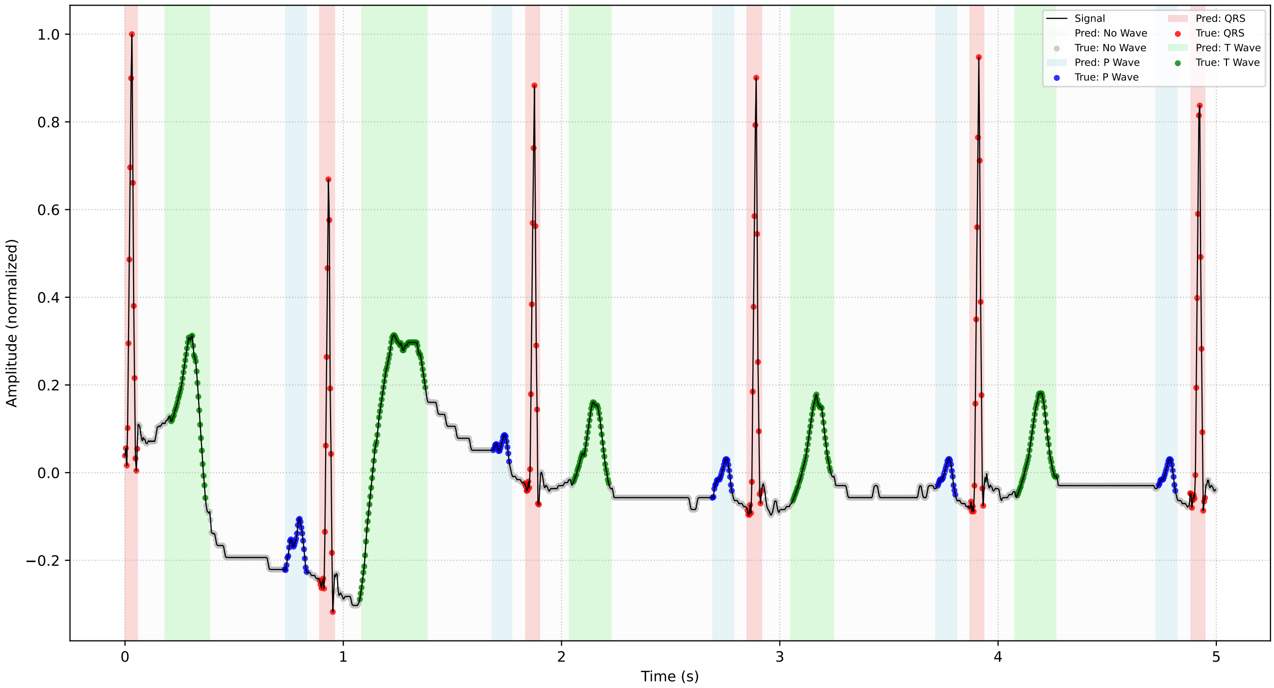


Figure 3.5 Segmentation performance of the U-Net-1D-900k model on a LUDB sample from the test set. This figure illustrates the type of sharp, high-frequency artifacts present in some LUDB recordings, which can pose a challenge for generalizing to MCG signals.

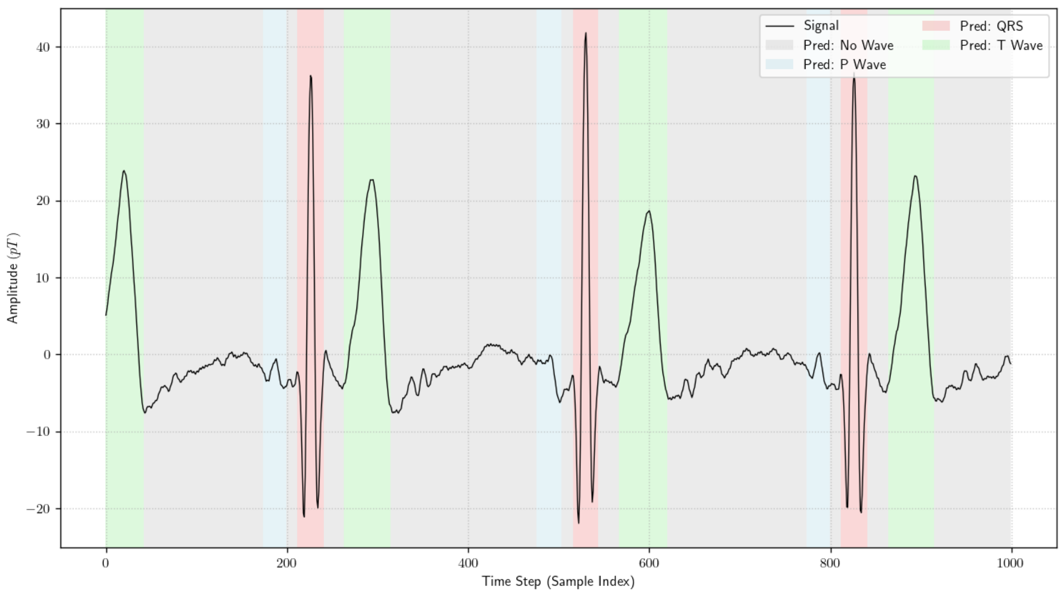


Figure 3.6 Segmentation of a filtered single-channel MCG signal using the trained U-Net-1D-15M model. Despite the rather high remaining noise level, the model successfully delineates the P-wave, QRS complex, and T-wave.

Furthermore, the 900k-parameter model performs only marginally worse than its 15M-parameter counterpart and other much larger models, like the one proposed by Joung et al. [15]. This demonstrates its high efficiency and suitability for deployment on resource-constrained infrastructure. Future improvements will thus depend largely on addressing the scarcity of large, high-quality, and consistently annotated open source datasets.

For all subsequent analyses in this work, the higher-performing 15M-parameter model is employed.

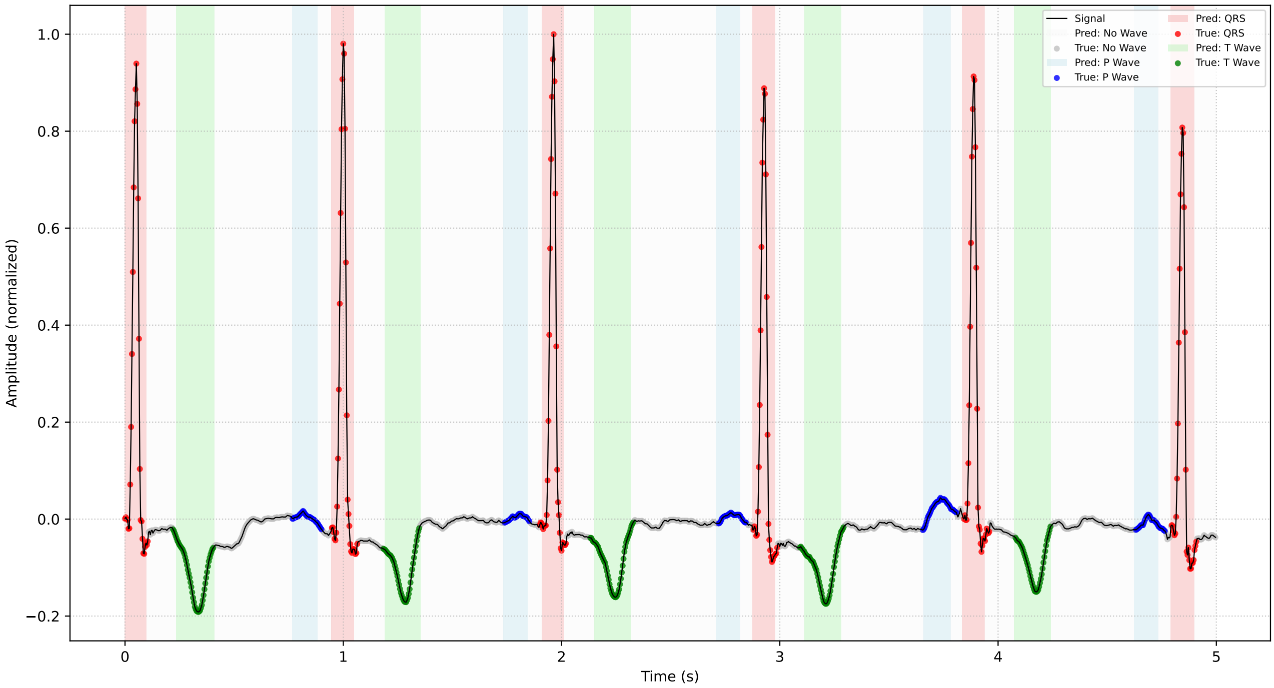


Figure 3.7 Example segmentation performance on a random sample from the test dataset. The model correctly identifies the P-wave, QRS complex, and an inverted T-wave, demonstrating its robustness to pathological signals.

3.4 Model Deployment and Practical Applications

Beyond its primary function as a segmentation tool, the trained U-Net model can be leveraged for several practical downstream applications that enhance the analysis of cardiac signals. Two such applications are detailed below: robust peak detection and a novel signal quality assessment score.

Robust Peak Detection The high F1-score of the model's QRS complex and T-wave detection (>98%, as shown in Table 3.2) enables a more robust peak detection methodology. By constraining the search for the signal's maximum absolute value to only those intervals identified as QRS or T segments, the algorithm's reliability is significantly improved. This approach is crucial for the precise timing of cardiac events, particularly in noisy MCG recordings. Figure 3.8 illustrates this process for R-peak detection.

A Novel "Heartbeat Score" for Signal Quality Assessment For multi-channel recordings such as MCG, it can often be beneficial to automatically identify the channel with the cleanest signal to perform further analysis. To automate this process, a novel *"Heartbeat Score"* is introduced. This metric quantifies the likelihood that a given signal segment represents a plausible cardiac rhythm by assigning it a score between 0 and 1. The score is calculated based on two main criteria:

- **Prediction Confidence:** The model's softmax output probabilities are aggregated to gauge the certainty of its classifications. Higher average probabilities suggest a more confident prediction and thus a higher quality signal.
- **Physiological Plausibility:** The relative proportions of the detected P, QRS, and T segments are compared against physiologically expected ranges for a typical heartbeat. Signals with segment distributions that deviate significantly from norms receive a lower score.

The final score combines these two components, providing a single metric to rank channels by signal quality. This concept is particularly useful for optimizing subsequent processing steps, such as the Independent Component Analysis (ICA) filtering explored in Chapter 4.1.

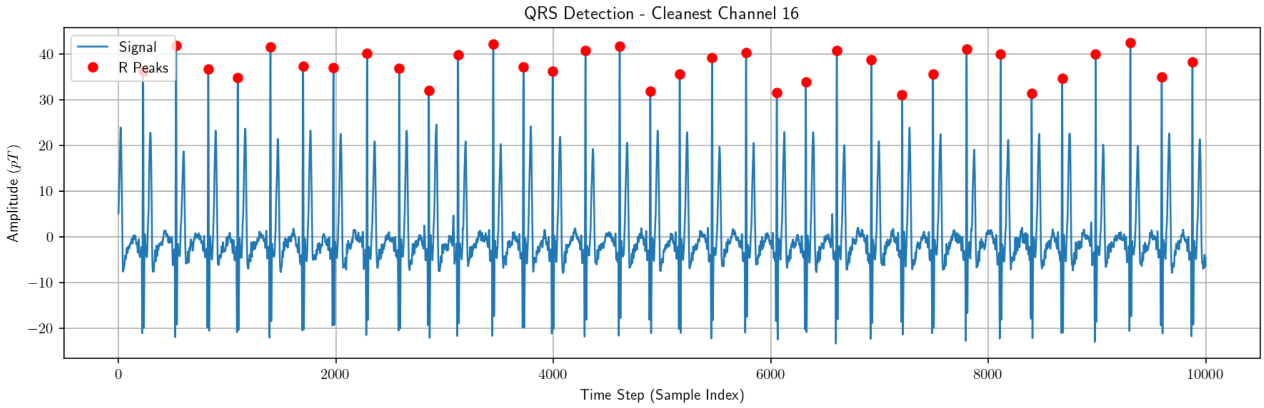


Figure 3.8 Illustration of robust R-peak detection on a pre-filtered MCG signal. The search for the peak is constrained to the interval identified by the model as the QRS complex, preventing erroneous detection of noise spikes.

4 Magnetocardiography to Screen Adults with ACM: A pilot study

This chapter is based on a pilot study conducted in collaboration with the German Heart Center, Munich. The findings are part of a manuscript in preparation for publication by Friese, Wacker-Gussmann et al.

The World Health Organization (WHO) defines primary cardiomyopathy as "cardiomyopathy associated with a mechanical or electrical dysfunction of the heart". One of these cardiomyopathies, which includes both dysfunctions, is arrhythmogenic cardiomyopathy (ACM). This preferentially affects the right ventricle (but also the left ventricle), in which heart muscle cells are replaced by fibrotic fatty tissue during the disease. This ventricle remodeling leads to the arrhythmogenicity that gives the disease its name [24]. The prevalence is estimated to be between 1:250 and 1:1000 [29], making it one of the more common of the rare diseases. The onset of the disease in adults is, on average, 31 years, but it can vary considerably (4-64 years) [24]. Clinical criteria have been established to determine the probability of ACM (so-called task force criteria). These criteria, which were revised in 2010, include a family history, abnormalities in imaging diagnostics (echocardiography, cardio-MRI), and rhythmological analyses (resting and long-term ECG). The clinical diagnosis of ACM is considered confirmed if two major criteria, one major and two minor criteria, or four minor criteria are present [23].

A major criterion is detecting a disease-causing variant in one of the known disease genes. These disease genes often code for desmosomal proteins that mediate cell-cell interaction. Variants are most frequently found in the PKP2, DSP, and DSG2 genes [24]. However, a pathogenic variant can only be detected in around 50% of patients with task force-positive ACM [31].

A phenotype-genotype correlation (statement about the course of the disease based on the detected variant) is only possible to a limited extent. Although patients with a pathogenic variant in DSP are more likely to have left ventricular involvement [2], the course of the disease can vary considerably within a family with the same pathogenic variant [41]. This makes counseling and treating patients and their family members challenging, especially when it comes to prognostic statements. This clinical problem highlights the need to develop additional clinical parameters beyond the existing ones, which can help better characterize the disease in individual patients. Magnetocardiography is the magnetic counterpart of the ECG, providing additional non-invasive electrophysiological information. Due to the low permeability of the tissue, the magnetic field is only slightly affected. It can offer complementary spatial information compared to the electrical potentials measured by ECG. Therefore, we investigated magnetic field measurements in ACM-positive patients and compared them to those of healthy controls. The recordings were made using an OPM-based MCG system and a person-sized magnetic shield, established at TUM University, Munich, Germany.

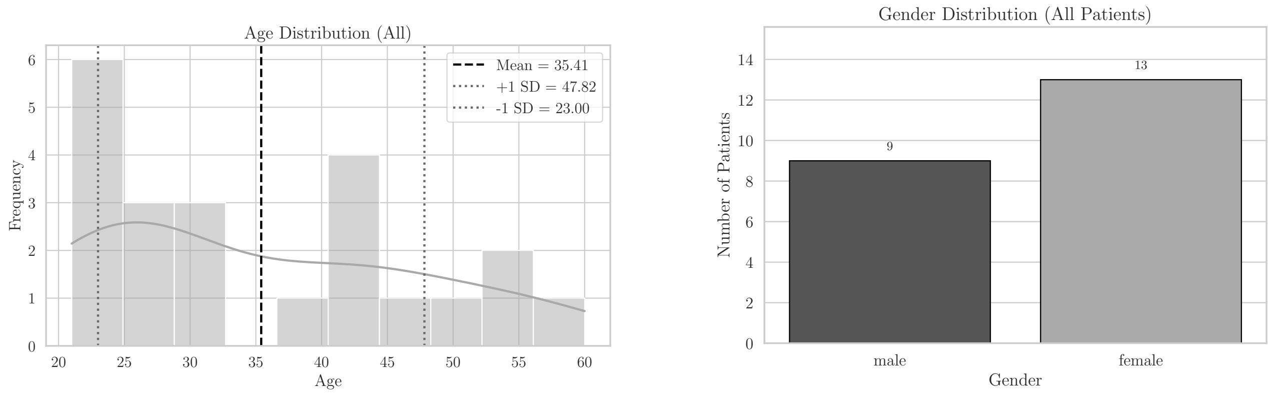
4.1 Methods

Subjects

The study cohort included participants with a diagnosis of ACM who presented at our university hospital, as well as healthy control subjects. The demographics of the cohort are detailed in Figure 4.1.

Inclusion criteria for ACM participants were a) Task force positive ACM patients with evidence of pathogenic variants (clinically and genetically confirmed ACM), or b) Task force positive ACM patients without evidence of a pathogenic variant (clinically confirmed ACM), or c) Task force negative carriers of a pathogenic variant (genetic predisposition to develop ACM).

Exclusion criteria were minors, participants incapable of giving consent, or the inability to take information due to a language barrier, and those with ICDs. Healthy volunteers were defined as individuals with no history of cardiac disease and an unremarkable physical examination.



(a) **(b)**

Figure 4.1 (a) Age distribution and **(b)** gender distribution of the study cohort, including both healthy individuals and ACM patients.

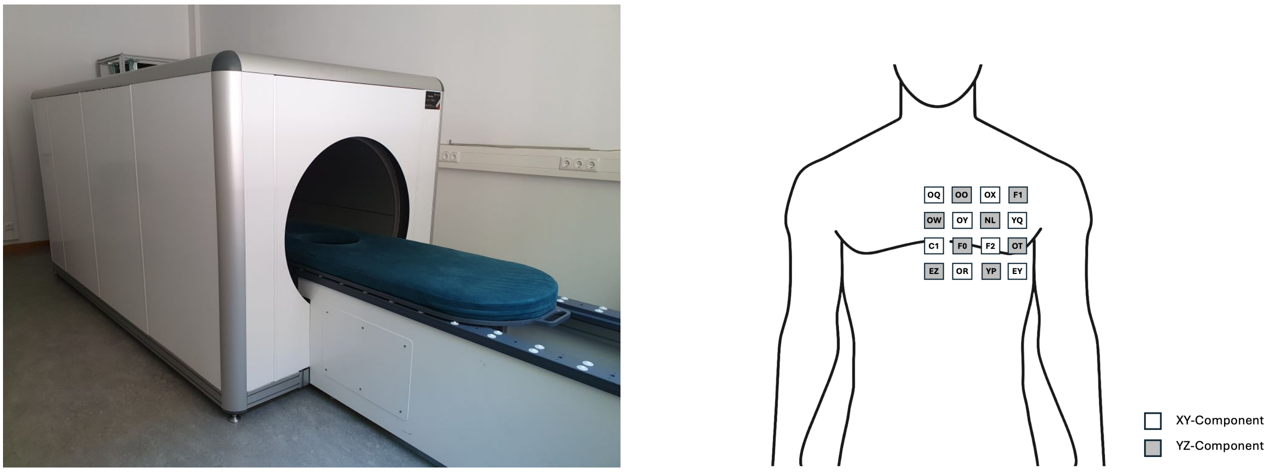
Magnetocardiography

Magnetocardiography (MCG) is a non-invasive imaging modality that maps the weak magnetic fields generated by cardiac electrical activity. While historically reliant on cryogenic Superconducting Quantum Interference Devices (SQUIDs), the recent advent of Optically Pumped Magnetometers (OPMs) has significantly reduced the cost and complexity of the technology, making clinical application more feasible [44]. The fundamental advantage of MCG over standard electrocardiography lies in the physical nature of the recorded signal and its interaction with the body. The ECG measures scalar surface potentials generated by passive, extracellular volume currents. These currents are driven through the body and are heavily dependent on the heterogeneous electrical conductivities of intervening tissues. High-impedance layers, such as subcutaneous fat or lung tissue, significantly attenuate and distort electrical potentials [36]. Conversely, the MCG is directly sensitive to the magnetic field generated by intracellular primary currents. Since biological tissues are magnetically transparent, the cardiac magnetic field propagates to the sensor array without the attenuation inherent to surface potentials. Crucially, this implies that observed changes in magnetic signal amplitude are a direct reflection of the underlying myocardial source strength. Consequently, multi-channel MCG arrays provide a high-fidelity spatial map of cardiac activity that complements the information resolved by the standard 12-lead ECG [44].

Measurement Setup and Data Acquisition

For the measurements, an advanced OPM system at the German Heart Center at the Technical University Munich was used. This setup, depicted in Figure 4.2a, is built around a cylindrical, person-sized magnetic shield (Magnetic Shields Limited) constructed from three layers of 1.5 mm thick MuMetal®. This passive shielding provides a noise floor of 220 $\mathrm{fT}/\sqrt{\mathrm{Hz}}$ in the frequency band of interest (3–45 Hz). The accessible interior, lined with polyvinyl chloride, has a diameter of 97 cm and a length of 265 cm. An additional current loop at the shield's opening provides passive compensation against static field influx [43]. Further noise reduction is achieved using a tri-axial Magnetic Field Cancelling System (MR-3, Stefan Mayer Instruments), which provides active field compensation and lowers the ambient noise floor to 80 fT/$\sqrt{\text{Hz}}$. This final noise level is about six times larger than the intrinsic noise floor of the magnetometers themselves [43].

The sensor array consists of sixteen QuSpin Zero Field Magnetometers (QZFMs) fixed in a flat 4 × 4 grid with 25.3 mm spacing to account for potential sensor crosstalk, as shown in Figure 4.2b. The grid is positioned below the participant's abdomen, with shimming to adjust for differences in body size. In this configuration, the surface of the abdomen is 1.5 mm from the sensor's surface and 8 mm from the sensitive volume of the magnetometer. To mitigate thermal irritation, a thin layer of air is passed between the sensor heads and the participant. The OPM sensors are operated in dual-axis mode to measure two orthogonal components of the magnetic field [43].



(a) **(b)**

Figure 4.2 (a) The measurement setup at the German Heart Center, showing the multi-layer magnetic shield and the patient table. **(b)** Layout of the 4×4 grid of dual-axis QuSpin sensors, including sensor IDs, magnetic field components measured, and position relative to the patient's torso.

Participants wore non-magnetic clothing. While both positions were initially considered, only data recorded in the prone position were included in the final analysis, as supine measurements demonstrated lower data quality. Each recording lasted at least five minutes. Although 21 participants were measured, sporadic sensor malfunctions necessitated the exclusion of specific channels. As a result, the number of valid observations for certain sensors fell below the total participant count ($N < 21$). The whole measurement took around 10 minutes.

Signal Processing and Averaging

Raw MCG data underwent a two-stage preprocessing pipeline. First, to isolate relevant cardiac frequencies, we applied a digital bandpass filter (1–95 Hz). Concurrently, a band-stop filter was utilized to attenuate 50 Hz powerline interference and its harmonics. Second, any baseline drift and DC offset were removed through third-order polynomial detrending. Following preprocessing, a lightweight neural network was employed to segment the continuous MCG signal. The network classified the signal into four distinct waveform classes: *no wave*, *P wave*, *QRS complex*, and *T wave*. (see Chapter 3)

To quantify the signal quality in each channel, we developed a composite score that integrates the neural network's performance with physiological constraints. This score for each channel i is formulated as a weighted sum of the network's mean confidence and a heartbeat plausibility score.

$$\text{Score}_i = \alpha \cdot \bar{C}_i + \beta \cdot P_i \tag{4.1}$$

The components of this score are defined as follows:

$\bar{C}_i$ is the **mean confidence** for channel i, averaged over T time points:

$$\bar{C}_i = \frac{1}{T}\sum_{t=1}^{T} C_{i,t}$$

P_i is the **plausibility score**. This score is derived from the deviation of the observed MCG segment proportions from ideal physiological ranges. A higher score indicates a more physiologically plausible waveform.

$$P_i = \frac{1}{1 + 0.1 \cdot D_i}$$

D_i is the **total deviation** from ideal segment durations, defined as the sum of the individual segment deviations:

$$D_i = d_{\mathsf{P},i} + d_{\mathsf{QRS},i} + d_{\mathsf{T},i}$$

(α, β) are the weighting parameters, set to default values of $(\alpha = 0.80)$ and $(\beta = 0.20)$.

The deviation for each segment ($d_{\mathsf{seg},i}$) is calculated based on how much its relative duration ($p_{\mathsf{seg},i}$) falls outside a predefined ideal range. Let $p_{\mathsf{seg},i}$ be the percentage of a given segment (P, QRS, or T) within a single heartbeat in channel i. The deviation is then:

$$d_{\mathsf{seg},i} = \max(0, L_{\mathsf{seg}} - p_{\mathsf{seg},i}) + \max(0, p_{\mathsf{seg},i} - U_{\mathsf{seg}}) \tag{4.2}$$

Here, L_{seg} and U_{seg} are the lower and upper bounds of the **ideal physiological range** for each segment:

- **P-wave:** [8%, 15%]
- **QRS complex:** [8%, 15%]
- **T-wave:** [15%, 30%]

The channel quality score plays a crucial role in guiding artifact removal via Independent Component Analysis (ICA). ICA was applied independently to each magnetic field component, using the respective sensor channels as input. The output components were then evaluated using the plausibility score. Components exhibiting high similarity to valid heartbeat morphology (i.e., a high plausibility score) were retained. Conversely, components identified as having low cardiac relevance were discarded. The final, cleaned signal was then reconstructed using the ICA Mixing Matrix ($\mathbf{A}$) (see Chapter 2.3) and only the retained components. This method allows us to preserve the essential vectorial information of the cardiac signal while effectively removing noise sources. An example of this filtering is shown in Figure 4.3.

For QRS detection, we selected the single channel with the highest quality score from the full set of sensors and their orthogonal components. QRS complexes were identified within this optimal channel over intervals spanning at least 30 seconds. Subsequently, to enhance the signal-to-noise ratio and suppress statistical fluctuations, average waveform segments were computed for each channel. These segments were centered around the detected QRS peaks using a windowed averaging approach, as shown in Figure 4.4.

Measurements

All subsequent analyses were performed on the average MCG waveforms. We assumed that, due to the proximity of the sensors to the heart, cardiac activity was recorded simultaneously across the entire sensor grid. The boundaries of the P wave, QRS complex, and T wave of the averaged waveforms were initially estimated by the neural network and subsequently refined through manual review. A relatively large timing uncertainty, modeled as a Gaussian distribution centered around 0 ms with a standard deviation of ±40 ms, was deliberately assumed for each manually adjusted segment boundary to rule out potential biases from

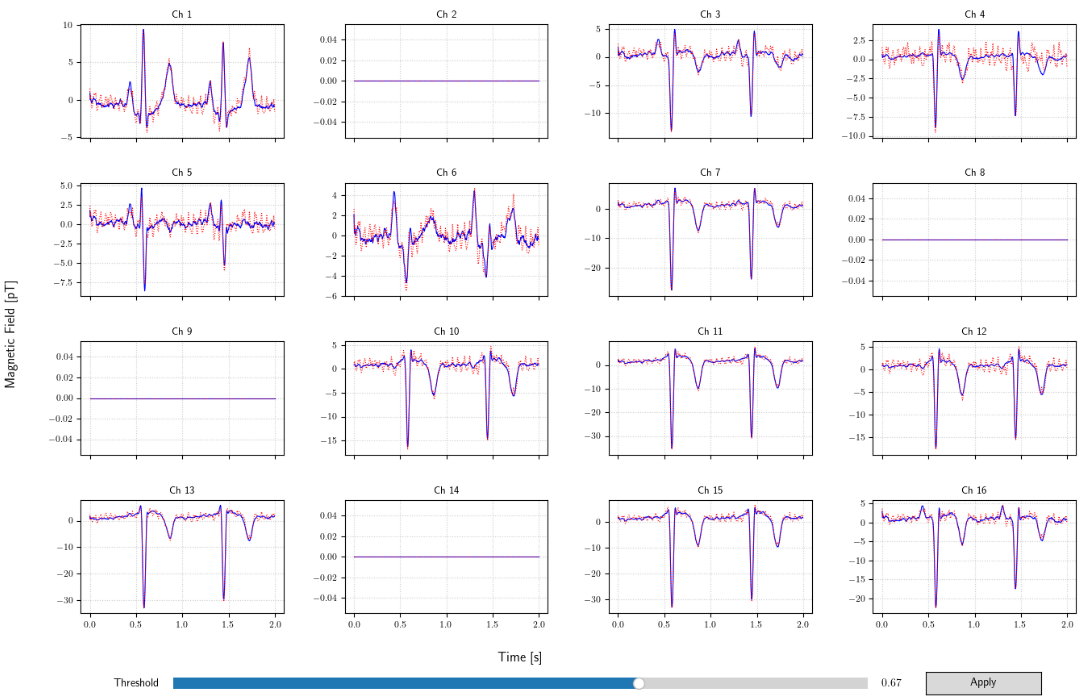


Figure 4.3 Exemplary result of the ICA filtering approach applied to the y-component that underwent prior bandpass/bandstop filtering. In this example, four sensors were excluded due to a malfunction. The red dotted line represents the data before ICA filtering, while the blue line corresponds to the data after filtering. Essential signal features are preserved, while noise artifacts are largely removed. (The sensor grid is displayed from the patient's perspective.)

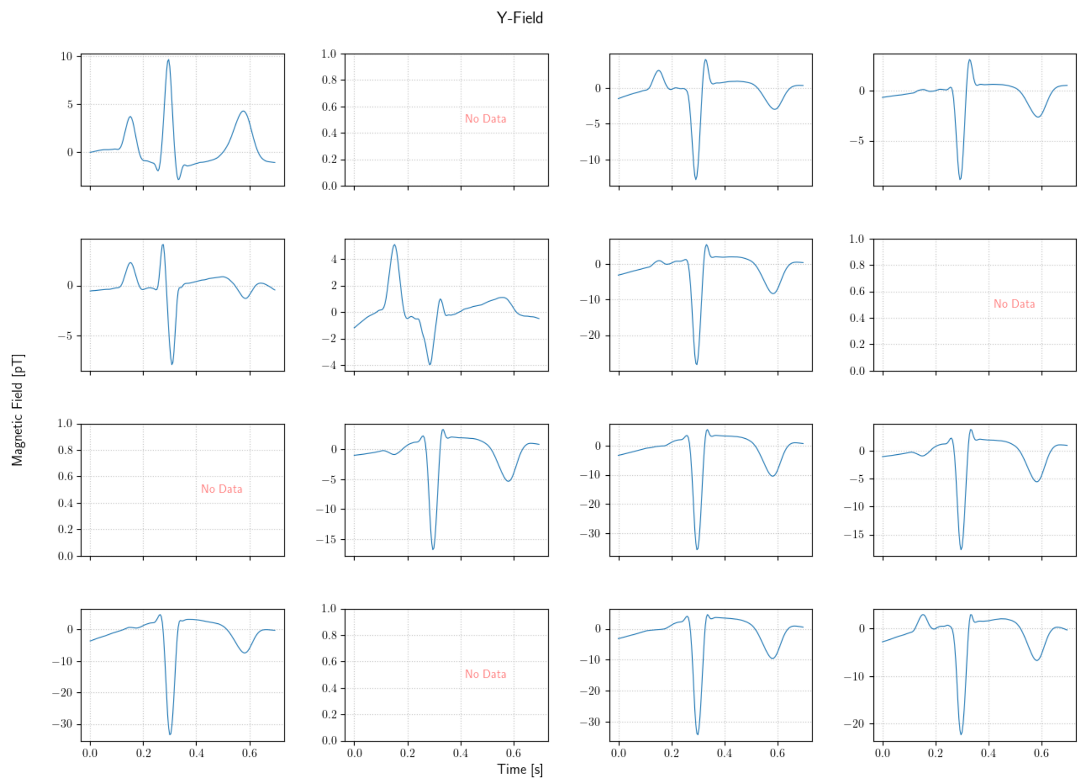


Figure 4.4 Exemplary result of windowed averaging after ICA filtering and QRS peak detection, shown for the same patient as in Figure 4.3.

differing segmentations. Feature extraction was subsequently performed on three key cardiac intervals defined by these boundaries: the full QRS complex, the ST segment (spanning from the J-point to the onset of the T-wave), and the complete T-wave.

For each of the three defined segments and every sensor, we constructed a two-dimensional vector magnetocardiogram (VMCG) (see Figure 1.1): For each sensor, the time-varying magnetic vector $\mathbf{B}(t)$ was projected onto the xy and yz planes (depending on the components they measured, 4.6), creating a trajectory loop. Unlike scalar ECG intervals, these loops capture both the spatial and temporal evolution of the depolarization and repolarization wavefronts.

1. **Maximum Magnetic Field Amplitude ("Distance")**

   Definition: The maximum Euclidean distance from the loop's origin to its furthest point.

   Physiological Significance: This proxies the peak magnitude of intracellular currents. A reduced distance implies signal attenuation, potentially due to the loss of viable myocardium (fibro-fatty replacement) characteristic of ACM.

2. **Convex Hull Area (Total Spatial Dispersion)**

   Definition: The area of the smallest convex polygon enclosing the vector trajectory (Figure 4.5, dashed boundary).

   Physiological Significance: This metric captures the magnitude and asymmetry of the magnetic field vector throughout a segment (such as the T wave). A larger area signifies a distinct divergence between the initial and terminal phases, whereas a minimal area characterizes either signal attenuation or a symmetric return to baseline.

3. **Net Area (Signed Area)**

   Definition & Significance: Calculated via the Shoelace formula, this signed metric accounts for loop directionality (Figure 4.5, gray shaded area). Unlike Hull Area, self-intersecting loops (e.g., figure-8 patterns) result in the mathematical cancellation of positive and negative areas, distinguishing open loops from complex, fragmented patterns.

4. **Shape Descriptors (Solidity & Compactness)**
   Two dimensionless ratios assessed morphology independent of amplitude:

   - **Solidity** (Net Area/Hull Area): Quantifies loop coherence. Values near 1.0 indicate open loops; low values identify twisting or self-intersections (Figure 4.5).
   - **Compactness** ($4\pi \cdot$Area/Perimeter^2): Quantifies circularity. High values indicate regular, round shapes; low values imply elongated or irregular trajectories.

Statistical Analysis

Statistical comparisons of magnetic heart vector features were performed between ACM subjects and healthy controls, across three cardiac segments (QRS complex, ST segment, and T wave) and three geometric metrics (enclosed area, distance, compactness).

Group differences for each segment–metric combination were assessed using Welch's two-sample t-test, where a 2-tailed p-value < 0.05 was deemed statistically significant. Diagnostic performance was evaluated using receiver operating characteristic (ROC) curve analysis, with optimal thresholds determined by maximizing Youden's index. Sensitivity, specificity, F_1-score (see Chapter 3.3), and area under the curve (AUC) were reported for each threshold.

To evaluate the robustness of our findings against measurement imprecision, specifically the uncertainty in defining the exact start and end times of cardiac segments, we incorporated measurement uncertainty via

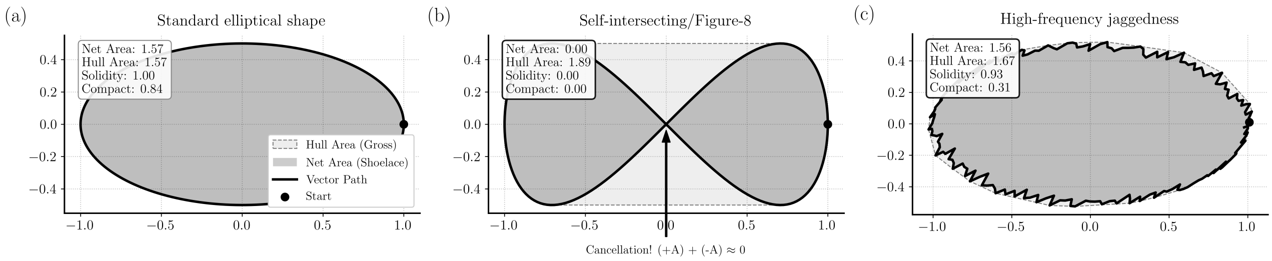
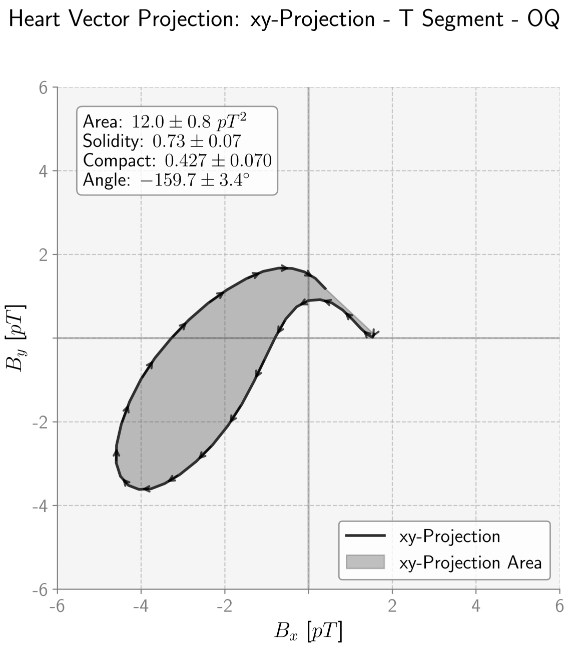
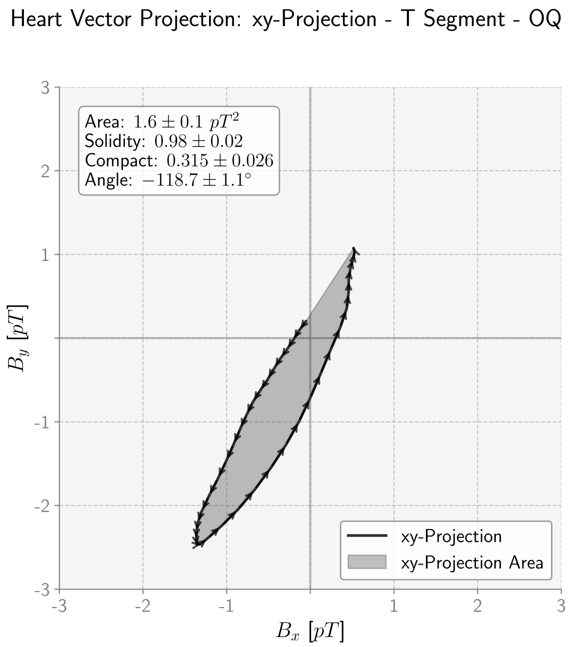


Figure 4.5 Comparison of Net Area (signed polygon; dark gray fill) and Convex Hull Area (light gray dashed) across three scenarios. **(a)** Standard Elliptical Shape: Represents a coherent trajectory where Net and Hull areas are identical, yielding a Solidity (Net / Hull) of ≈ 1.0 and high Compactness (not 1 as the shape isn't a perfect circle). **(b)** Self-Intersecting Loop: A "figure-8" trajectory where the positive and negative signed components of the Net Area mathematically cancel out (approaching 0). The Hull Area remains unaffected, capturing the full spatial extent, which results in a distinctively low solidity and compactness score. **(c)** Fragmentation: A trajectory characterized by jagged path deviations. While the general convex shape is preserved (Solidity ≈ 1.0), the increased perimeter length significantly reduces the Compactness score ($4\pi \cdot$ Area/Perimeter^2).



(a)



(b)

Figure 4.6 Example VMCG projections during the T-wave segment, along with calculated metrics for the OQ sensor (see Figure 4.2b), for **(a)** a healthy participant and **(b)** a patient diagnosed with ACM. (Note that axes are scaled differently) a Monte Carlo simulation with $N = 5,000$ iterations. It is important to note that this approach differs from bootstrapping methods used to address small sample sizes; rather, this simulation serves as a sensitivity analysis to ensure that significant differences are not artifacts of specific segmentation choices. In each iteration, synthetic datasets were sampled from normal distributions centered at the nominal metric values, with standard deviations derived from the measurement uncertainty (corresponding to the modeled ± 40 ms segmentation tolerance). Statistical tests and threshold determinations were repeated on these iterations to estimate 95% uncertainty intervals (UI) for the statistical parameters, including p-values. The result was considered highly robust only if the entire 95% UI for the p-value remained below the 0.05 significance level, indicating that the finding persists even when accounting for measurement error. The UI reflects the range in which the considered metric would fall in 95 out of 100 repeated experiments. The entire statistical analysis was performed using Python version 3.11.2.

4.2 Results

This prospective feasibility study enrolled 21 participants, comprising 9 patients diagnosed with ACM and 12 healthy controls. The study was conducted as basic research. The mean age of the ACM cohort was 41 (SD $\pm$ 14) years, and all diagnoses were evaluated by a referring cardiologist. The healthy controls had a mean age of 31 (SD $\pm$ 9) years; none had a history of congenital or acquired cardiac disease or were taking any medication. The baseline characteristics of the ACM patients are shown in Table 4.1.

Table 4.1 Table 1: Clinical, genetic, and imaging characteristics of the ACM study cohort. (Abbreviations: ACM: Arrhythmogenic Cardiomyopathy; TFC: Task Force Criteria; MRI: Magnetic Resonance Imaging; RVEF: Right Ventricular Ejection Fraction; VES: Ventricular Extrasystoles; BBB: Bundle Branch Block.)

Pt	Age	Sex	TFC Maj	TFC Min	TFC Pos	Genetic Findings	MRI Findings	Echo Findings	ECG Findings	Fam. Hist
1	53	F	2	1	Yes	c.2014-1G>C, PKP2 het, (NM_001005242.2), pathogenic	RV dilated, dyskinesia anterolateral and RV, interseptal microaneurysmata, RVEF 35%	Hypo- and akinesia of the apical free wall	> 2500 VES in 24h	Neg
2	39	M	2	0	Yes	Negative	Transmural fibrotic area lateral RV midventricular, thinning of myocardium, motility disorder, RVEF 34%	–	Negative T wave in III, aVR and V1	Pos
3	46	M	2	0	Yes	Negative	Late gadolinium enhancement, septal RV	VES during measurement, otherwise normal	VES, complete BBB, VT	Pos
4	60	F	2	0	Yes	c.2014-1G>C, PKP2 het, (NM_001005242.2), pathogenic	n.a.	RV apex aneurysm	VT	Pos
5	21	F	2	0	Yes	c.2014-1G>C, PKP2 het, (NM_001005242.2), pathogenic	RV normal, RVEF 65%, no regional bulging, LVEF 71%	Dyskinesia	Sinus rhythm, no pathological findings	Pos
6	56	F	2	0	Yes	Test pending	Circumscribed hypokinesia midventricular inferolateral wall, LVEF low-normal, basal inferior fibrosis	Normal	Sinus rhythm, no pathological findings	Pos
7	30	M	1	2	Yes	Negative	RVEF 38%, hypokinesia apical RV	Normal	Late potential positive	Pos
8	52	F	1	2	Yes	c.369G>A, p.(Trp123*), PKP2 het, (NM_001005242.2), pathogenic	Small aneurysms RV (RVOT and inferior RV wall), RVEF 53%	Mild RV bulging, RV dyskinesia, RV function normal	Sinus rhythm, no pathological findings	Pos
9	35	F	1	2	Yes	Mutation in DSP*	RVEF 41%, late enhancement posterolateral LV	Global hypokinesia, EF 31%	AV block I$^{\circ}$, VES (< 500/24h)	Neg

The primary finding of this study is a statistically significant reduction in the VMCG loop dimensions in ACM patients compared to healthy controls. This was most pronounced during ventricular repolarization (T-wave) and, to a lesser extent, during depolarization (QRS complex). Specifically, ACM patients exhibited significantly lower values for Convex Hull Area (total dispersion) and Distance (maximum amplitude) (Figures 4.7 and 5). In contrast, no significant differences were found in the ST segment. Notably, the shape descriptors: Solidity (Twist Index) and Compactness, did not differ significantly between groups. This suggests that while the ACM signal is attenuated (due to a smaller loop area and amplitude), the topology of the loop remains coherent and does not exhibit increased twisting or fragmentation in these 2D projections compared to the controls.

Analysis of the T-wave Convex Hull Area revealed significant differences in 14 of 16 sensors. Spatially, these differences were concentrated in the lower and right-sided regions of the sensor array, corresponding to the anatomical position of the ventricles (Figure 4.9). Crucially, for 13 of these 14 sensors, the entire 95% uncertainty interval (UI) for the p-value remained below the 0.05 threshold (Table 4.2). This confirms that the distinction in convex hull area persists even when accounting for segmentation uncertainty. A similar, though slightly less widespread, reduction was observed in the T-wave Distance metric (significant in 10 of 16 sensors). Additionally, the diagnostic thresholds appear to be higher in the yz projections compared to the xy projections (Figure 4.7).

Table 4.2 Mean p-values and 95% uncertainty intervals for the T-wave convex hull area metric across the sensors that showed significant differences

Sensor	Mean p-value	95% UI Lower	95% UI Upper
F0	0.0036	0.0010	0.0089
OR	0.0271	0.0123	0.0479
EZ	0.0110	0.0019	0.0393
OX	0.0171	0.0063	0.0361
OW	0.0021	0.0008	0.0045
OO	0.0108	0.0053	0.0190
C1	0.0015	0.0001	0.0060
OY	0.0067	0.0018	0.0174
F2	0.0196	0.0099	0.0355
EY	0.0463	0.0171	0.0917
YP	0.0142	0.0026	0.0339
NL	0.0269	0.0125	0.0491
OT	0.0106	0.0037	0.0236
OQ	0.0015	0.0004	0.0042

In addition to repolarization abnormalities, our analysis revealed distinct alterations during ventricular depolarization (QRS complex). Most notably, the QRS Distance (maximum amplitude) was significantly reduced in 12 of 16 individual sensors. Eight of these sensors demonstrated statistical robustness (upper 95% UI < 0.05; Table 4.3), indicating a widespread loss of electromotive force. The QRS Convex Hull Area was also significantly smaller in 8 of 16 sensors.

Table 4.3 Mean p-values and 95% uncertainty intervals for the QRS-wave distance metric across the sensors that showed significant differences

Sensor	Mean p-value	95% UI Lower	95% UI Upper
F0	0.0190	0.0049	0.0484
OR	0.0106	0.0042	0.0233
F1	0.0467	0.0170	0.1017
OW	0.0063	0.0018	0.0141
OO	0.0034	0.0004	0.0124
C1	0.0108	0.0023	0.0319
OY	0.0093	0.0015	0.0287
F2	0.0016	0.0007	0.0037
YP	0.0334	0.0113	0.0783
NL	0.0011	0.0001	0.0042
YQ	0.0241	0.0058	0.0640
OQ	0.0385	0.0038	0.1451

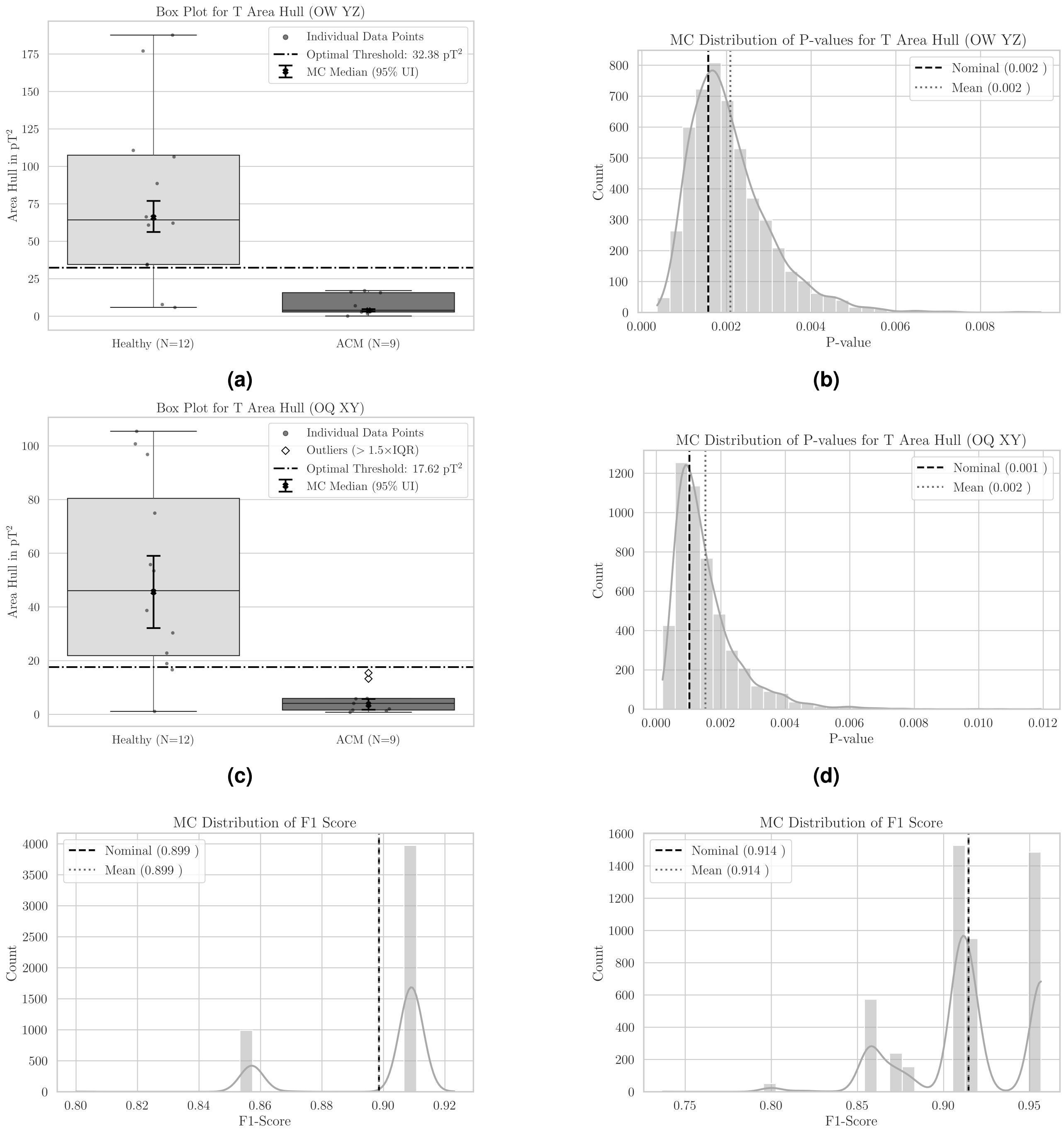


Figure 4.7 T-wave area analysis for sensors OW and OQ (see Figure 4.2b). **(a, c)** Boxplots of nominal VMCG area values (in pT^2). The shaded boxes represent the interquartile range (IQR), while the bold error bars denote 95% Monte Carlo–based uncertainty intervals for the median. These intervals represent the range within which the median is expected to fall with 95% probability, given the simulated measurement uncertainties. The nominal median is indicated by a horizontal line. Note that sensor OW **(a)** recorded the *yz*-projection and sensor OQ **(c)** recorded the *xy*-projection. **(b, d)** Distributions of p-values from the Monte Carlo simulations for OW and OQ, respectively. **(e, f)** Distributions of F1 scores from the same simulations for OW and OQ, indicating classification performance (see Chapter 3.3).

To mitigate spatial uncertainty arising from anatomical variability and array positioning, we performed an aggregated sub-grid analysis. The 4×4 array was partitioned into four 2×2 quadrants, and vector components within each quadrant were averaged (e.g., combining adjacent sensors in the bottom-right corner). The statistical analysis was then repeated on these averaged datasets. The results are shown in Tables 4.4, 4.5 and Figure 4.8. As in the individual sensor analysis, the most significant effects were observed in the lower and right half of the grid.

Aggregated analysis of the T-wave Convex Hull Area corroborated the findings from the single-sensor analysis. With the exception of the top left (yz) component, all quadrants demonstrated robust statistical significance (95% UI < 0.05), as shown in Table 4.4. The strongest effects were identified in the right-sided quadrants. Specifically, the bottom right (yz) component showed the lowest mean p-value (0.0015; 95% UI: 0.0004–0.0043), followed closely by the top right (xy) component ($p = 0.0023$; 95% UI: 0.0006–0.0064).

Table 4.4 Mean p-values and 95% uncertainty intervals (UI) for the aggregated analysis of T-wave Convex Hull Area across significant quadrants of the sensor array (from the patient's perspective).

Source	Mean p-value	95% UI Lower	95% UI Upper
Top Left (yz)	0.0195	0.0037	0.0625
Top Right (xy)	0.0023	0.0006	0.0064
Top Right (yz)	0.0042	0.0017	0.0083
Bottom Left (xy)	0.0253	0.0106	0.0496
Bottom Left (yz)	0.0103	0.0021	0.0257
Bottom Right (xy)	0.0085	0.0023	0.0207
Bottom Right (yz)	0.0015	0.0004	0.0043

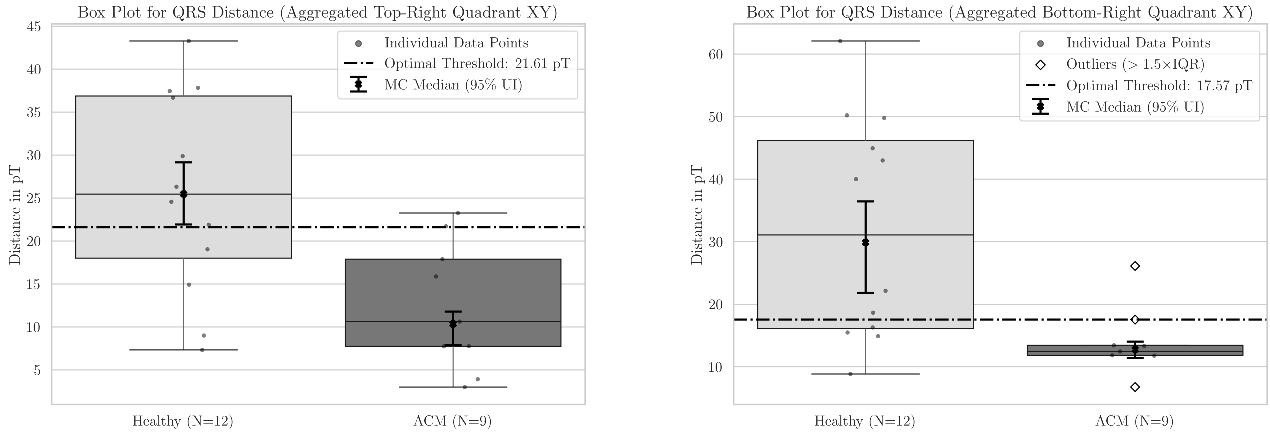
The aggregated analysis of the QRS Distance (maximum amplitude) also revealed significant signal attenuation in ACM patients (Table 4.5, Figure 4.8). Robust differences (UI < 0.05) were identified across multiple quadrants, including the top left, top right, and bottom left. However, the top right (yz) component exhibited the strongest effect ($p = 0.0029$; 95% UI: 0.0005–0.0092).

Table 4.5 Mean p-values and 95% uncertainty intervals (UI) for the aggregated analysis of the QRS-complex distance metric across significant quadrants of the sensor array (from the patient's perspective).

Source	Mean p-value	95% UI Lower	95% UI Upper
Top Left (yz)	0.0052	0.0011	0.0146
Top Right (xy)	0.0109	0.0013	0.0397
Top Right (yz)	0.0029	0.0005	0.0092
Bottom Left (xy)	0.0183	0.0089	0.0354
Bottom Right (xy)	0.0074	0.0024	0.0185
Bottom Right (yz)	0.0366	0.0086	0.0985

4.3 Discussion

The primary finding of this study is a significant reduction in the magnetic field strength of ACM patients compared to healthy controls, as evidenced by smaller VMCG loop areas and shorter distances. This signal attenuation was most pronounced during the T-wave (repolarization) but was also confirmed in the QRS complex. These results align well with the underlying pathophysiology of ACM, where functional cardiomyocytes are progressively replaced by fibro-fatty tissue. As the mass of viable myocardium decreases, the magnitude of the generated intracellular currents, and consequently the resulting magnetic fields, diminishes. The fact that shape factors remained stable lends support to this hypothesis. It implies that



(a) **(b)**

Figure 4.8 Boxplots following the same layout as Figure 4.7, showing aggregated results for the xy-component of the QRS distance metric in the **(a)** top-right sensors and **(b)** bottom-right sensors.

although the quantity of active myocardium is reduced, the spatial coherence of the remaining electrical activity appears to be largely maintained.

Crucially, this direct correlation between tissue loss and signal reduction is difficult to establish reliably using standard surface ECG. ECG potentials are heavily attenuated and distorted by patient-specific factors, such as thoracic geometry, lung volume, and varying distributions of subcutaneous fat, which obscure the true strength of the cardiac source [36]. In contrast, the cardiac magnetic field permeates biological tissue largely unhindered [44]. Therefore, the reduced field strength observed in our ACM cohort could serve as a direct, physical reflection of the structural myocardial loss, unconfounded by body habitus. Furthermore, these effects were predominantly localized to the lower and right-sided sensor regions (from the patient's perspective, Figure 4.9), corresponding to the anatomical position of the right ventricle.

We speculated that measuring the heart's magnetic field could supplement the established Task Force criteria and help screen at-risk individuals. At this stage, we cannot yet determine if MCG might also serve as an early disease indicator, especially in diagnostically ambiguous cases; further studies with more participants are needed to establish this.

There are several issues to consider. In ECG analysis, late potentials are one criterion for ACM. However, the necessary data filtering to remove background noise in MCG recordings also eliminates the low-frequency signals characteristic of late potentials.

Although not in regular use today, vectorcardiography was previously used to diagnose ACM. It had limitations in detecting normal ECG findings and various borderline or abnormal findings such as T-wave inversions and pathological Q waves [34]. However, some small studies reported that vectorcardiography could demonstrate varying degrees of conduction delay in the right ventricle, showing a localized right pre-cordial QRS prolongation as a highly specific finding in ACM subjects [29].

MRI is often considered a gold standard imaging technique for detecting fatty tissue infiltration in the right ventricles of patients with typical ACM. However, these fatty infiltrations are not always demonstrable. A larger study comparing MCG with MRI findings is certainly warranted, but we speculate that MCG may offer more precise findings related to the electrophysiological consequences of the disease.

Recent MCG studies by other groups have already shown its potential as a diagnostic tool in inflammatory cardiomyopathies [11]. For example, one study described a high specificity (95%) with a relatively low sensitivity (59%); however, the low sensitivity was primarily attributed to the participants' immunosuppressive or anti-inflammatory medication already being taken. In untreated patients, the sensitivity was reportedly significantly higher [4]. In another case of cardiomyopathy related to wild-type amyloidosis, the response to tafamidis treatment was successfully monitored using VMCG analysis [8]. These studies align with our experience.

In the prospective MagMa study, Heidecker et al. included 110 patients with chest pain after exclusion of obstructive coronary heart disease and 220 matched healthy individuals. They reported high accuracy for MCG, with a specificity of 98%. In three cases, the MCG procedure was even more accurate than MRI. In contrast to the MagMa study, which used a superconducting quantum interference device (SQUID), our study used optically pumped magnetometers. This newer technology allows for a wide, more patient-comfortable, and much more cost-effective device than SQUID systems [43].

Overall, little has been reported in the literature about the magnetic field of the heart in subjects with ACM. The biomagnetic pattern we found in patients with ACM is of particular interest and requires further evaluation. The expressivity of the genes within a family is variable. The reduced penetrance is around 60%, meaning that not all carriers will develop ACM. VMCG may help identify early signs in genotype-positive patients who have not yet exhibited any clinical symptoms.

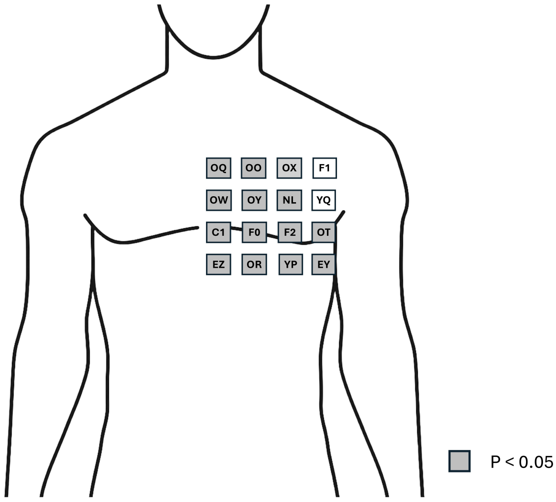


Figure 4.9 Map of the 4x4 sensor grid (patient's perspective) showing sensors (in gray) with a (average) statistically significant difference ($p < 0.05$) in the T-wave VMCG area between healthy subjects and those with ACM.

Limitations of the Study

A key limitation of this study is the low number of cases that could ultimately be included. This was primarily due to excessive noise in data from patients with older ICDs, which often rendered the recordings uninterpretable. The strong magnetic fields produced by older ICD models introduced substantial electromagnetic interference (EMI) into the MCG recordings. In nearly every case, this interference generated a magnetic field (> 50 nT) that exceeded the operational dynamic range of the field-nulling algorithm within the Optically Pumped Magnetometers (OPMs) [28] (see Chapter 2.2). This issue dramatically decreased the sensitivity of the instruments. This high noise level also obscured subtle cardiac signals, resulting in a low signal-to-noise ratio that rendered the data unreliable for interpretation. Consequently, patients with these older ICDs were necessarily excluded from the final analysis, contributing to the smaller number of cases included in the study. Technical improvements are underway to address these issues, as data from newer devices is more readily analyzable. Also, measurements taken in better shielded environments, such as magnetically shielded rooms (MSRs), require significantly less filtering, which may further enhance data quality in future analyses. Nevertheless, given the promising results of this pilot study, further evaluations will be conducted.

5 Conclusion and Outlook

5.1 Conclusion and Outlook

Conclusion

This thesis demonstrated the potential of a machine learning-enhanced magnetocardiography (MCG) methodology for screening Arrhythmogenic Cardiomyopathy (ACM). To enable a robust analysis of complex cardiac signals, particularly from noisy, short-interval Optically Pumped Magnetometer (OPM) recordings, a U-Net based deep learning model was developed for automated waveform segmentation. This tool provided the foundation for applying advanced signal analysis to the challenging data acquired in the clinical portion of this work.

The subsequent pilot study, analyzing data from 9 ACM patients and 12 healthy controls, revealed this work's principal finding: a statistically significant reduction in the VMCG loop dimensions (both convex hull area and maximum amplitude) in patients with ACM. This signal attenuation was most pronounced during the T-wave repolarization phase, spatially concentrated over the lower and right-sided regions of the sensor array. Furthermore, significant reductions in the QRS complex (depolarization) were confirmed through aggregated sub-grid analysis. These results suggest that VMCG provides a sensitive, physical marker for the loss of viable myocardium characteristic of ACM, holding promise as a non-invasive tool to supplement existing diagnostic criteria.

While the conclusions drawn from this pilot study must be interpreted with caution, given the primary limitation of a small cohort size and technical challenges with electromagnetic interference from older Implantable Cardioverter-Defibrillators (ICDs), this thesis has successfully demonstrated a proof-of-concept. The findings lay the groundwork for a promising new approach in cardiac diagnostics, offering a patient-comfortable and cost-effective alternative to SQUID-based systems.

Outlook

The findings of this thesis open several promising avenues for future research. The most critical next step is to validate these results in a larger and more diverse patient cohort. Such a study would not only establish more reliable sensitivity and specificity metrics but could also permit longitudinal analysis. Of particular interest is the screening of genotype-positive family members who have not yet developed the phenotype; VMCG could potentially detect early electrophysiological changes before they manifest in standard imaging or ECG.

Beyond this specific project, the U-Net segmentation model developed here has significant potential. Its proven robustness on noisy, short-interval signals makes it an excellent candidate for other demanding applications, such as fetal magnetocardiography (fMCG). Applying the model to fMCG data could automate the delineation of fetal heartbeats, a task that is critical for the non-invasive prenatal diagnosis of cardiac conditions.

Finally, to better understand the underlying signal changes that cause the observed differences in the VMCG, future studies should be conducted in an environment with lower magnetic noise, such as a magnetically shielded room (MSR). The cleaner data from an MSR would require less aggressive signal filtering. This could preserve subtle, low-frequency features of the cardiac signal—specifically the late potentials that are a known diagnostic criterion for ACM but were necessarily removed by the filters in this study. Analyzing such high-quality data would allow for a more detailed examination of the cardiac magnetic field and could help uncover new diagnostic markers for the disease.

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